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ROBOT SYSTEM, CONTROL METHOD FOR ROBOT SYSTEM, MANUFACTURING METHOD FOR ARTICLE USING ROBOT SYSTEM, INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, PROGRAM, AND RECORDING MEDIUM

Abstract

A robot system includes a robot and a controller that controls the robot. The controller, when a user is moving a predetermined part of the robot or moving the predetermined part with a terminal device, notifies the user that the predetermined part is moving from a first position to a second position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of International Patent Application No. PCT/JP2023/041486, filed Nov. 17, 2023, which claims the benefit of Japanese Patent Application No. 2022-190044, filed Nov. 29, 2022, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

Field of the Technology

[0002] The present disclosure relates to a robot.

Description of the Related Art

[0003] In recent years, collaborative robots capable of performing collaborative work with operators have been developed in the field of industrial robots. This collaborative robot allows an operator to perform direct teaching, in which the operator teaches the robot by directly touching the robot to change the posture of the robot. With this direct teaching, a user is able to intuitively operate the robot, so it is possible to easily perform teaching. PTL 1 describes a method for a robot to move in plot units during direct teaching by providing discrete, virtual three-dimensional plots in the working space of the robot for the purpose of improving the operability of robot operation by the user.

CITATION LIST

Patent Literature

[0004] PTL 1 PCT Japanese Translation Patent Publication No. 2018-532608

[0005] However, PTL 1 does not discuss how to notify the user of the amount of movement of the robot when the user operates the robot. By notifying the user of the amount of movement of the robot, it is possible to further improve the operability of robot operation by the user.

SUMMARY

[0006] The present disclosure aims to improve the operability of robot operation by a user.

[0007] In the present disclosure, a robot system includes a robot and a controller that controls the robot, wherein the controller, when a user is moving a predetermined part of the robot, notifies the user that the predetermined part is moving from a first position to a second position.

[0008] Features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic diagram of a robot system **1000** in an embodiment.

[0010] FIG. 2 is a control block diagram of the robot system **1000** in the embodiment.

[0011] FIG. 3A is a diagram showing a virtual coordinate system and a virtual reference coordinate **3001** of a working space of the robot system **1000** in the embodiment.

[0012] FIG. 3B is a diagram for illustrating virtual repulsion in the embodiment.

[0013] FIG. 3C is a diagram for illustrating virtual attraction in the embodiment.

[0014] FIG. 4A is a diagram for illustrating a method of notifying a user by a change in operating force in the embodiment.

[0015] FIG. 4B is a diagram for illustrating a method of notifying a user by a change in operating

force in the embodiment.

[0016] FIG. 5 is a control flowchart in the embodiment.

[0017] FIG. 6A is a diagram for illustrating a method of notifying a user by a change in operating force in an embodiment.

[0018] FIG. 6B is a diagram for illustrating a method of notifying a user by a change in operating force in the embodiment.

[0019] FIG. 6C is a diagram for illustrating a method of notifying a user by a change in operating force in the embodiment.

[0020] FIG. 7 is a control flowchart in the embodiment.

[0021] FIG. 8 is a diagram for illustrating a method of notifying a user by a change in operating force in the embodiment.

[0022] FIG. 9 is a diagram for illustrating a method of notifying a user by a change in operating force in an embodiment.

[0023] FIG. 10 is a diagram for illustrating a method of notifying a user by a change in operating force in an embodiment.

[0024] FIG. 11 is a control flowchart in the embodiment.

[0025] FIG. 12 is a diagram for illustrating a method of notifying a user by a change in operating force in an embodiment.

[0026] FIG. 13 is a diagram for illustrating a method of notifying a user by a change in operating force in the embodiment.

[0027] FIG. 14 is a control flowchart in the embodiment.

[0028] FIG. 15 is a diagram for illustrating a method of notifying a user by light in an embodiment.

[0029] FIG. 16 is a control flowchart in the embodiment.

[0030] FIG. 17 is a diagram for illustrating a method of notifying a user by light in the embodiment.

[0031] FIG. 18 is a control flowchart in the embodiment.

[0032] FIG. 19 is a diagram for illustrating a method of notifying a user by sound in an embodiment.

[0033] FIG. 20 is a control flowchart in the embodiment.

[0034] FIG. 21 is a diagram for illustrating a method of notifying a user by sound in the embodiment.

[0035] FIG. 22 is a control flowchart in the embodiment.

[0036] FIG. 23 is a diagram for illustrating a method of notifying a user by sound in the embodiment.

[0037] FIG. 24 is a diagram for illustrating a method of notifying a user by sound in an embodiment.

[0038] FIG. 25 is a control flowchart in the embodiment.

[0039] FIG. 26 is an example of a direct teaching settings screen **800** in an embodiment.

[0040] FIG. 27 is a diagram for illustrating a method of notifying a user by vibration in an embodiment.

[0041] FIG. 28 is a diagram for illustrating a method of notifying a user by an external input device **500** in an embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0042] Hereinafter, embodiments of the present disclosure will be described with reference to the attached drawings. The embodiments described below are only illustrative, and, for example, the configuration of detailed parts may be modified as needed by persons skilled in the art without departing from the purport of the present disclosure. Numeric values described in the present embodiments are reference numeric values and do not limit the present disclosure. In the following drawings, the arrows X, Y, Z in the drawings represent the overall coordinate system of a robot system. Generally, an XYZ three-dimensional coordinate system represents a world coordinate

system of an overall installation environment. Other than that, for the sake of convenience of control, a local coordinate system may be used as needed for a robot hand, a finger part, a joint, or the like.

First Embodiment

[0043] FIG. 1 is a schematic diagram of the robot system **1000** in the present embodiment when viewed in a selected direction in an XYZ coordinate system. As shown in FIG. 1, the robot system **1000** includes an articulated robot arm body **200** and a robot hand body **300** as a robot body. The robot system **1000** further includes a controller **400** that controls the operation of the entire robot body. The robot system **1000** also includes an external input device **500** as a teaching device that transmits teaching data to the controller **400**. One example of the external input device **500** includes a teaching pendant. The external input device **500** is used by an operator to designate the position of the robot arm body **200** and the position of the robot hand body **300**.

[0044] In the present embodiment, the description will be made when a robot hand is provided at the distal end part of the robot arm body **200** as an end effector; however, a device provided at the distal end part of the robot arm body **200** is not limited thereto, and the device may be a tool or the like.

[0045] A link **201** that is the proximal end of the robot arm body **200** is provided at a base **210**. The robot arm body **200** includes a base **210**, a plurality of joints **J1** to **J6**, for example, six joints (six axes), and a plurality of links **201** to **205**. The robot arm body **200** further includes a plurality of (six) arm motors **211** to **216** at the joints **J1** to **J6** as drive sources that respectively rotationally drive the joints around their rotation axes. Each of the arm motors **211** to **216** includes a motor encoder (not shown) that detects the rotational position of a motor output shaft.

[0046] The robot arm body **200** further includes arm motor controllers **221** to **226** for respectively controlling the arm motors **211** to **216**, and includes force sensors **251** to **256** (FIG. 2) each of which detects torque as information about a force acting on a corresponding one of the joints. For the sake of simplicity, in FIG. 1, the arm motor controllers **221** to **226** are shown outside the robot arm body **200**; however, it is assumed that the arm motor controllers **221** to **226** are respectively provided near the corresponding arm motors **211** to **216** inside the base **210** and the links **201** to **205**.

[0047] In the robot arm body **200**, the plurality of links **201** to **205** and the robot hand body **300** are rotatably connected at the joints **J1** to **J6**. Here, the links **201** to **205** are sequentially connected in series from the proximal end to the distal end of the robot arm body **200**.

[0048] As shown in the diagram, the base **210** and link **201** of the robot arm body **200** are connected by the joint **J1** that rotates in the direction of the arrow around the X-axis in the diagram. The link **201** can receive the rotation of the arm motor **211**, transmitted through a transmission mechanism (not shown), and rotate in the direction of the arrow around the Z-axis in the diagram.

[0049] The link **201** and link **202** of the robot arm body **200** are connected by the joint **J2** that rotates in the direction of the arrow around the Y-axis in the diagram. The link **202** can receive the rotation of the arm motor **212**, transmitted through a transmission mechanism (not shown), and rotate in the direction of the arrow around the Y-axis in the diagram.

[0050] The link **202** and link **203** of the robot arm body **200** are connected by the joint **J3** that rotates in the direction of the arrow around the Y-axis in the diagram. The link **203** can receive the rotation of the arm motor **213**, transmitted through a transmission mechanism (not shown), and rotate in the direction of the arrow around the Y-axis in the diagram.

[0051] The link **203** and link **204** of the robot arm body **200** are connected by the joint **J4** that rotates in the direction of the arrow around a predetermined axis positioned in the XZ plane in the diagram. The link **204** can receive the rotation of the arm motor **214**, transmitted through a transmission mechanism (not shown), and rotate in the direction of the arrow around the predetermined axis positioned in the XZ plane in the diagram.

[0052] The link **204** and link **205** of the robot arm body **200** are connected by the joint **J5** that

rotates in the direction of the arrow around the Y-axis in the diagram. The link **205** can receive the rotation of the arm motor **215**, transmitted through a transmission mechanism (not shown), and rotate in the direction of the arrow around the Y-axis in the diagram.

[0053] The link **205** of the robot arm body **200** and the robot hand body **300** are connected by the joint **J6** that rotates in the direction of the arrow around a predetermined axis positioned in the XZ plane in the diagram. The robot hand body **300** can receive the rotation of the arm motor **216**, transmitted through a transmission mechanism (not shown), and rotate in the direction of the arrow around the predetermined axis positioned in the XZ plane in the diagram.

[0054] The robot hand body **300** is used to grip an object, such as a component and a tool. The robot hand body **300** of the present embodiment opens and closes two finger parts **312** with a drive mechanism (not shown) and a hand motor **311** to grip or release a workpiece, and grips a workpiece so as not to displace the workpiece relative to the robot arm body **200**.

[0055] The robot hand body **300** includes a built-in hand motor controller (not shown) for controlling the drive of the hand motor **311**. The robot hand body **300** is connected to the link **205** via the joint **J6**. When the joint **J6** rotates, the robot hand body **300** can also be rotated.

[0056] Each of the arm motor controllers **221** to **226** is communicably connected to the controller **400** by a communication line **103**. The communication line **103** is a communication line that connects the controller **400** to each of the arm motor controllers **221** to **226** and that is used for communicating instructions from the controller **400** and replies from the arm motor controllers **221** to **226**.

[0057] The controller **400** transmits controlled target values for the arm motors **211** to **216** to the arm motor controllers **221** to **226** based on motion trajectories and the like input in advance from the external input device **500**, to integrate and control the arm motor controllers **221** to **226**. Each of the arm motor controllers **221** to **226** transmits various information such as the current angle of a corresponding one of the arm motors **211** to **216** to the controller **400**. The transmission from the controller **400** to each of the arm motor controllers **221** to **226** and the transmission from each of the arm motor controllers **221** to **226** to the controller **400** are performed at a predetermined communication cycle. The connection method between the controller **400** and each of the arm motor controllers **221** to **226** may be a cascade connection, a bus connection, or a daisy chain connection. In the present embodiment, a bus connection is used for description.

[0058] Here, the hand part of the robot arm body **200** is the robot hand body **300** in the present embodiment. When the robot hand body **300** is gripping an object, the hand part of the robot arm body **200** includes the robot hand body **300** and the object being gripped (such as a component and a tool). In other words, regardless of whether the robot hand body **300** is in a state of gripping an object or in a state of gripping no object, the robot hand body **300** that is an end effector is referred to as a hand part.

[0059] With the above configuration, the robot hand body **300** can be moved by the robot arm body **200** to a selected position and be caused to perform a desired task. For example, it is possible to manufacture an assembled workpiece as a product by using a predetermined workpiece and a different workpiece as materials and performing a process of assembling the predetermined workpiece and the different workpiece. As a result, it is possible to manufacture an article with the robot arm body **200**.

[0060] The robot hand body **300** may be an end effector or the like, such as a pneumatically-driven air hand. The robot hand body **300** is assumed to be attached to the link **205** by a semi-fixed means, such as screw fastening, or can be attached by an attaching/detaching means, such as latch fastening. Particularly, when the robot hand body **300** is detachably attachable, a system in which the robot arm body **200** is controlled to detach and attach or replace multiple types of robot hand bodies **300** placed at a supply position through the action of the robot arm body **200** itself is also conceivable.

[0061] FIG. **2** is a control block diagram of the robot arm body **200**. The arm motor controllers **221**

to **226** respectively provided at the joints of the robot arm body **200** respectively include motor drivers **231** to **236** and CPUs (central processing units) **241** to **246**, and are respectively connected to corresponding force sensors **251** to **256**.

[0062] The controller **400** includes a CPU **401**, and an ROM (read only memory) **402**, an RAM (random access memory) **403**, an HDD (hard disk drive) **404**, and a recording disk drive **405** as a storage portion. The controller **400** includes an input/output interface (not shown) for communicating with the external input device **500**. The CPU **401**, the ROM **402**, the RAM **403**, the HDD **404**, and the recording disk drive **405** are communicably connected by a bus **406**.

Furthermore, the communication line **103** is connected to the CPU **401** so that the CPU **401** can communicate with each of the CPUs **241** to **246** of the joints.

[0063] The ROM **402** is a non-transitory storage device. The ROM **402** stores a basic program **450**, such as BIOS, that is used to cause the CPU **401** to execute various arithmetic processing and that is read by the CPU **401** during start-up time. The CPU **401** executes various arithmetic processing in accordance with the basic program recorded on (stored on) the ROM **402**. The basic program **450** can be stored on the HDD **404**.

[0064] The RAM **403** is a temporary storage device used in arithmetic processing of the CPU **401**. The HDD **404** is a non-transitory storage device that stores various data, such as arithmetic processing results of the CPU **401**. The recording disk drive **405** is capable of reading various data, programs, and the like, recorded on a recording disk **440**.

[0065] In the present embodiment, the basic program **450** is recorded on the ROM **402**; however, the configuration is not limited thereto. The basic program **450** may be recorded on any recording medium as long as the recording medium is a non-transitory computer-readable recording medium. Examples of the recording medium for supplying the basic program **450** to a computer include a flexible disk, an optical disk, a magneto-optical disc, a magnetic tape, and a nonvolatile memory.

[0066] Each of the CPUs **241** to **246** is a CPU that controls a corresponding one of the arm motors **211** to **216** in accordance with an instruction from the CPU **401** of the controller **400**. Each of the force sensors **251** to **256** is a sensor that periodically detects the force applied to a corresponding one of the joints **J1** to **J6** and that outputs the detection result to a corresponding one of the control CPUs **241** to **246**. In the present embodiment, a torque sensor is used to detect the torque applied to each joint as a force, but the sensor is not limited thereto. If the sensor is capable of acquiring information about a disturbance on the robot arm body **200**, any type of sensor may be used. Each of the motor drivers **231** to **236** is a driver circuit that generates current to control a corresponding one of the arm motors **211** to **216** based on the input signal of a corresponding one of the control CPUs **241** to **246**.

[0067] The CPU **401** receives teaching point data input from, for example, the external input device **500** via an interface (not shown). The CPU **401** is also capable of generating the trajectory of each axis of the robot arm body **200** based on the teaching point data input from the external input device **500** and transmitting the trajectory to a corresponding one of the CPUs **241** to **245** via the communication line **103**. The CPU **401** outputs data of a drive instruction indicating the controlled amount of the rotation angle of each of the arm motors **211** to **216** to a corresponding one of the CPUs **241** to **246** at predetermined intervals.

[0068] Each of the CPUs **241** to **246** calculates the amount of current output to a corresponding one of the arm motors **211** to **216** based on the drive instruction received from the CPU **401** and outputs the calculated amount of current to a corresponding one of the motor drivers **231** to **236**. Each of the motor drivers **231** to **236** supplies current to a corresponding one of the arm motors **211** to **216** to control the joint angle of a corresponding one of the joints **J1** to **J6**. Each of the CPUs **241** to **246** executes feedback control of a corresponding one of the arm motors **211** to **216** such that the current joint angle value of a corresponding one of the joints **J1** to **J6**, acquired based on the rotation angle of the motor output shaft, detected by a motor encoder (not shown), becomes a target joint angle.

[0069] Each of the force sensors **251** to **256** outputs information related to a force, which is a detection result, to a corresponding one of the CPUs **241** to **246**. In other words, each of the CPUs **241** to **246** is capable of acquiring a change in torque applied to a corresponding one of the joints **J1** to **J6** and detected by a corresponding one of the force sensors **251** to **256**. It is also possible to perform force control and stop the robot arm body **200** based on the information about the force from the force sensors **251** to **256**. Based on the detection results from these force sensors **251** to **256**, it is possible to perform direct teaching to control and teach the robot based on the forces generated by the user directly operating the robot arm body **200**.

[0070] Here, the basic control of direct teaching according to the present embodiment will be briefly described. In the present embodiment, impedance control, which is a type of force control, will be described as an example of a method of executing direct teaching. Impedance control means control in a manner such that there are a virtual spring and a virtual damper between the hand part position (current position) and target position of the robot arm body **200** and the hand part of the robot arm body **200** is operated as if the force based on the size of the spring and damper is generated at the hand part of the robot arm body **200**.

[0071] Where D denotes a damping coefficient, K denotes a spring coefficient, x denotes a current position, and x_d denotes a target position (fulcrum of the spring), a target force (the force to be generated at the hand part of the robot) F_d is expressed as follows.

$$[00001] D(x(\cdot) - x_d(\cdot)) + K(x - x_d) = F_d$$

[0072] In direct teaching, the target position x_d is set based on, for example, the magnitude of force applied by human operation, and the target force F_d is determined according to the damping coefficient D and the spring coefficient K . In short, in the present embodiment, the damping coefficient D and the spring coefficient K in impedance control are used as force control parameters in the present embodiment. However, using the damping coefficient D and the spring coefficient K as force control parameters as in the case of the first embodiment is just one example and does not limit the force control parameters to these parameters. In the present embodiment, in addition to the force applied by the user in the above equation, virtual repulsion and attraction are taken into consideration to change the damping coefficient D and the spring coefficient K , thereby changing the operating force needed during direct teaching. The operating force may translate to a resistance force or resistance when the hand part of the robot arm body **200** is moved during direct teaching.

[0073] Next, the details of direct teaching in the present embodiment will be described in detail. FIGS. **3A**, **3B**, and **3C** are diagrams for illustrating direct teaching in the present embodiment. A link **205** or the robot hand body **300** may be referred to as a predetermined part. FIG. **3A** is a diagram illustrating a virtual coordinate system and a virtual reference coordinate **3001** in a working space of the robot system **1000**. FIG. **3B** is a diagram where a repulsion field is set in the virtual coordinate system. FIG. **3C** is a diagram where an attraction field is set in the virtual coordinate system. FIGS. **3A**, **3B**, and **3C** simply show the robot arm body **200** and the robot hand body **300** in the robot system **1000** described with reference to FIG. **1**. The virtual force field is assumed to act at the reference coordinate **3001** and does not act at coordinates other than the reference coordinate **3001** in the robot arm body **200** and the robot hand body **300**.

[0074] Here, the ordinate axis of each of FIGS. **3A**, **3B**, and **3C** represents a force [N] indicating the magnitude of a force detection value at the hand part of the robot arm body **200**, acquired by the force sensors **251** to **256**. The abscissa axis is X [m] indicating the position on the X -axis of the hand part of the robot arm body **200**. In the present embodiment, an X -axis in one axis direction is set as the virtual coordinate system set in the robot system **1000**. The X -axis of the upper diagram in each of FIGS. **3A**, **3B**, and **3C** is the X -axis of the coordinate system virtually set in the robot system **1000**, and the lower diagram in each of FIGS. **3A**, **3B**, and **3C** is the X -axis of the coordinate system related to the virtual force field, corresponding to the X -axis. The F -axis is an axis that indicates the strength and direction of a virtual force, showing the strength of repulsion

when the force is on the upper side and the strength of attraction when the force is on the lower side. In teaching of a robot, coordinate systems, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate **3001** is moved in the coordinate system of the entire robot system **1000** will be described as an example.

[0075] From FIG. 3A, in direct teaching, the movable range of the robot arm body **200** is limited to a predetermined direction, and furthermore, the user is notified when the hand part of the robot arm body **200** has moved a desired distance. In the present embodiment, the movable range is limited (restricted) to the X-axis direction of the robot system **1000**. The reference coordinate **3001** is a reference coordinate used as a reference when virtual repulsion or attraction is acquired. When the reference coordinate **3001** is in proximity to a coordinate, at which virtual repulsion or attraction is generated, as a result of operation of the user, the operating force needed for the user to operate the hand part of the robot arm body **200** changes. The operating force changes by controlling virtual resistance at the time when the user moves the hand part of the robot arm body **200** according to the position of the reference coordinate **3001** and the virtual force field. The virtual force field is considered on the assumption that the working space is a potential field and there are specific coordinates where repulsion or attraction is generated. When those coordinates are in close proximity to the reference coordinate **3001**, the operating force needed for the user to operate the robot arm body **200** changes under the influence of virtual repulsion or virtual attraction. These coordinates can be referred to as first positions or second positions.

[0076] From FIG. 3B, when the reference coordinate **3001** moves in the +X-axis direction and approaches a virtual repulsion coordinate **3002** where a virtual repulsion potential field **3003** having a virtual repulsion range **3004** is set, the operating force increases due to the virtual repulsion potential field **3003**. When the reference coordinate **3001** moves away from the virtual repulsion coordinate **3002**, the operating force reduces due to the virtual repulsion potential field **3003**. The virtual repulsion potential field **3003** is a range related to the magnitude of virtual repulsion, and is set larger as it approaches the virtual repulsion coordinate **3002**. The virtual repulsion range **3004** is a range where virtual repulsion is generated on the X-axis, and is centered on the virtual repulsion coordinate **3002**. Actually, virtual repulsion affects the entire working space of the robot system **1000**, but it is assumed that as it gets away from the virtual repulsion range **3004** and moves away from the virtual repulsion coordinate **3002**, virtual repulsion based on the virtual repulsion potential field **3003** reduces to an ignorable level.

[0077] Here, in the present embodiment, the virtual repulsion potential field **3003** and the virtual repulsion range **3004** are changed by the losses of mechanisms such as a speed reducer in the robot arm body **200**. Alternatively, a certain range set in advance may be constantly set. As the losses of mechanisms such as a speed reducer increase, the virtual repulsion potential field **3003** and the virtual repulsion range **3004** are reduced, with the result that it is possible to reduce a complete stop of the robot arm body **200** due to a combination of the losses of mechanisms and virtual repulsion.

[0078] Thus, when the reference coordinate **3001** approaches and passes through the virtual repulsion coordinate **3002** by the user operating the hand part of the robot arm body **200**, a force pushing from the virtual repulsion coordinate **3002** to the reference coordinate **3001** in the -X-axis direction is gradually applied, and the operating force gradually increases. When the reference coordinate **3001** passes through and moves away from the virtual repulsion coordinate **3002** by the user operating the hand part of the robot arm body **200**, a force pushing from the virtual repulsion coordinate **3002** to the reference coordinate **3001** in the +X-axis direction is significantly applied, and the operating force steeply reduces.

[0079] In this way, the operating force is changed to gradually increase when approaching the virtual repulsion coordinate **3002** and steeply reduce once passing through the virtual repulsion coordinate **3002**. By doing this, the user is able to obtain a clicking sensation at a coordinate, where

virtual repulsion is set, due to a change in operating force needed to operate the hand part of the robot arm body **200**. Thus, it is possible to notify the user of the passage of the virtual repulsion coordinate **3002** through the clicking sensation due to the change in operating force, so it is possible to grasp, through the clicking sensation, how much the robot arm body **200** has been moved.

[0080] In the present embodiment, when the operator stops direct teaching (stops operation) while the reference coordinate **3001** is present in the virtual repulsion range **3004**, the reference coordinate **3001** is controlled to be pushed from the inside of the virtual repulsion range **3004** to the outside of the virtual repulsion range **3004**. In other words, the reference coordinate **3001** is controlled to be positioned at the end of the virtual repulsion range **3004**.

[0081] From FIG. 3C, when the reference coordinate **3001** moves in the +X-axis direction and approaches a virtual attraction coordinate **3005** where a virtual attraction potential field **3006** having a virtual attraction range **3007** is set, the operating force reduces due to the virtual attraction potential field **3006**. When the reference coordinate **3001** moves away from the virtual attraction coordinate **3005**, the operating force increases due to the virtual attraction potential field **3006**. The virtual attraction potential field **3006** is a range related to the magnitude of virtual attraction, and is set larger as it approaches the virtual attraction coordinate **3005**. The virtual attraction range **3007** is a range where virtual attraction is generated on the X-axis, and is centered on the virtual attraction coordinate **3005**. Actually, virtual attraction affects the entire working space of the robot system **1000**, but it is assumed that as it gets away from the virtual attraction range **3007** and moves away from the virtual attraction coordinate **3005**, virtual attraction based on the virtual attraction potential field **3006** reduces to an ignorable level.

[0082] Here, in the present embodiment, the virtual attraction potential field **3006** and the virtual attraction range **3007** are changed by the losses of mechanisms such as a speed reducer in the robot arm body **200**. Alternatively, a certain range set in advance may be constantly set. As the losses of mechanisms such as a speed reducer increase, the virtual attraction potential field **3006** and the virtual attraction range **3007** are reduced, with the result that it is possible to reduce a complete stop of the robot arm body **200** due to a combination of the losses of mechanisms and virtual attraction.

[0083] Thus, when the reference coordinate **3001** approaches and passes through the virtual attraction coordinate **3005** by the user operating the hand part of the robot arm body **200**, a force pulling from the virtual attraction coordinate **3005** to the reference coordinate **3001** in the +X-axis direction is gradually applied, and the operating force gradually reduces. When the reference coordinate **3001** passes through and moves away from the virtual attraction coordinate **3005** by the user operating the hand part of the robot arm body **200**, a force pulling from the virtual attraction coordinate **3005** to the reference coordinate **3001** in the -X-axis direction is significantly applied, and the operating force steeply increases. In this way, the operating force is changed so as to gradually reduce when approaching the virtual attraction coordinate **3005** and steeply increase once passing through the virtual attraction coordinate **3005**.

[0084] In the present embodiment, when the operator stops direct teaching (stops operation) while the reference coordinate **3001** is in the virtual attraction range **3007**, the reference coordinate **3001** is controlled to be pulled to the virtual attraction coordinate **3005**. In other words, the reference coordinate **3001** is controlled so as to be positioned at the virtual attraction coordinate **3005**.

[0085] Next, direct teaching in the case where a plurality of virtual repulsion coordinates or a plurality of virtual attraction coordinates is set will be described with reference to FIGS. 4A and 4B. FIG. 4A is a diagram when virtual attraction coordinates are set at equal intervals in the X-axis direction. FIG. 4B is a diagram in which virtual attraction coordinates are placed at equal intervals in the X-axis direction and virtual repulsion coordinates are further placed outside of the virtual attraction coordinates placed at equal intervals.

[0086] From FIG. 4A, virtual attraction coordinates **4001** to **4008** are set in the X-axis direction, and virtual attraction potential fields **4011** to **4018** are respectively set at the virtual attraction

coordinates **4001** to **4008**. The reference coordinate **3001** is at the virtual attraction coordinate **4005**. Consider the case of moving the reference coordinate **3001** in this state from the virtual attraction coordinate **4005** to the virtual attraction coordinate **4006**. In this case, the magnitude of the influence reverses and the operating force abruptly changes under the influence of the virtual attraction coordinate closer to the reference coordinate **3001** in the process in which the reference coordinate **3001** approaches from the virtual attraction coordinate **4005** to the virtual attraction coordinate **4006**. In other words, where attraction at each virtual attraction coordinate is considered as a reference, a pseudo virtual repulsion potential field is set between the virtual attraction coordinates. When the reference coordinate **3001** passes through the position where the magnitude of the influence reverses, the force pulling to the virtual attraction coordinate **4005** changes into the force pulling to the virtual attraction coordinate **4006**. In other words, the force pulling in the $-X$ -axis direction changes into the force pushing in the $+X$ -axis direction. Therefore, a user who is moving the hand part in the $+X$ -axis direction can feel an abrupt change in operating force to obtain a clicking sensation.

[0087] When the user feels the change in operating force, the user can grasp how many virtual attraction coordinates the reference coordinate **3001** has moved and can grasp how much the robot arm body **200** has moved. The notification of the present embodiment is to notify how much the robot arm body **200** has moved, through a change in operating force in conjunction with the movement of the position of the reference coordinate **3001**. Since a virtual attraction potential field is generated at each virtual attraction coordinate, even when the user releases his or her hand from the robot arm body **200**, the hand part of the robot arm body **200** is positioned at the virtual attraction coordinate. Therefore, it is possible to reduce the unnecessarily excessive movement of the hand part of the robot arm body **200** although the user does not intend to move the hand, so it is preferable.

[0088] In FIG. 4A, the virtual attraction potential fields **4012**, **4017** at the virtual attraction coordinates **4002**, **4007** are increased compared to the other virtual attraction potential fields. Thus, it is possible to make the user strongly perceive a clicking sensation for each of the five virtual attraction coordinates, and to make the user perceive the passage of the fifth virtual attraction coordinates. Therefore, it is possible to reduce the likelihood that the user erroneously counts the number of virtual attraction coordinates passed. In the present embodiment, five virtual attraction coordinates are applied; however, any number of virtual attraction coordinates may be set.

[0089] In FIG. 4B, a virtual repulsion coordinate **4021** is set on the $-X$ -axis side of the virtual attraction coordinate **4002**, and a virtual repulsion coordinate **4028** is set on the $+X$ -axis side of the virtual attraction coordinate **4007**. Virtual repulsion potential fields **4031**, **4038** are respectively set at the virtual repulsion coordinates **4021**, **4028**.

[0090] Thus, it is possible to suppress movement in the $-X$ -axis direction beyond the virtual repulsion coordinate **4021** by the user and also suppress movement in the $+X$ -axis direction beyond the virtual repulsion coordinate **4028** by the user. Thus, for example, if there is an obstacle or the like in the $+X$ -axis direction beyond the virtual repulsion coordinate **4028** and may be brought into contact with the robot, it is possible to make it difficult for the user to operate the robot by virtual repulsion, with the result that it is possible to reduce the contact between the robot and the obstacle. Such repulsion may also be used to, for example, notify the number of coordinates passed.

[0091] FIG. 5 is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. 5 is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual attraction coordinates (positions) are defined as stationary positions, and the positions where the magnitude of influence reverses are defined as notification positions. As a premise, it is assumed that virtual coordinates, virtual repulsion fields, virtual attraction fields, stationary positions, and notification positions are set in control of the robot system **1000**.

[0092] As shown in FIG. 5, initially, in step **S501**, sensor values are acquired from the encoders

(not shown) provided in the motors **211** to **216**, to acquire the current position of the reference coordinate **3001**. In the present embodiment, the positions of the links **201** to **205** are acquired by the motor encoders. Alternatively, the positions of the links **201** to **205** may be directly detected by using an output shaft ENC that directly detects the positions of the links **201** to **205** or the hand part of the robot arm body **200** may be directly detected by an image capturing apparatus or the like.

[0093] Subsequently, in step **S502**, the positions where a force is applied to the reference coordinate **3001** are acquired based on the current position acquired in step **S501** and the notification positions and stationary positions set in advance. In the present embodiment, information about virtual coordinates, notification positions, stationary positions, and virtual repulsion and attraction information on virtual repulsion and attraction set at those positions is stored in a simulator storing the robot system **1000** as a model. Then, the information about the positions where a force is applied to the reference coordinate **3001** (the notification positions and the stationary positions) is acquired from the simulator. In the present embodiment, a simulator is used. Alternatively, various formats of data storing information about virtual coordinates, notification positions, stationary positions, and virtual repulsion and attraction set at those positions may be used.

[0094] Subsequently, in step **S503**, the force that is currently applied to the reference coordinate **3001** is acquired based on the current position of the reference coordinate **3001** and the information about the notification positions and the stationary positions. Subsequently, in step **S504**, the force applied by the user to the hand part of the robot arm body **200** is acquired from the values of the force sensors **251** to **256**.

[0095] Subsequently, in step **S505**, the forces in the respective axis directions (XYZ) at the reference coordinate **3001** are acquired based on the values of the force sensors **251** to **256**. In the present embodiment, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired based on the values of the force sensors **251** to **256** and the link parameters of the robot arm body **200**. Then, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired as the forces in the respective axis directions (XYZ) at the reference coordinate **3001**. This is because the link **205** that is the distal end link as the hand part of the robot arm body **200** is easy for the user to perform direct teaching. Of course, when the robot hand body **300** is operated to perform direct teaching, the forces in the respective axis directions (XYZ) at a predetermined position in the robot hand body **300** may be acquired. In that case, the forces in the respective axis directions (XYZ) are acquired from the values of the force sensors **251** to **256**, the link parameters of the robot arm body **200**, and the shape parameters of the robot hand body **300**.

[0096] In the subsequent step **S506**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **3001** operates in the specific axis direction. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X-axis direction at the reference coordinate **3001**. Thus, in step **S506**, the gains of the forces in the Y direction and Z direction of the robot arm body **200** are set to zero in the XYZ coordinate system. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X direction. Alternatively, the hand part of the robot arm body **200** may be moved in the Y direction or may be moved in the Z direction.

[0097] In the subsequent step **S507**, the force generated at the reference coordinate **3001** is acquired based on the forces that are applied to the reference coordinate **3001** from the virtual repulsion potential fields and/or virtual attraction potential fields set at the notification positions and the stationary positions, and the force applied from the user. Then, in step **S508**, the damping coefficient **D** and the spring coefficient **K** are changed using the force generated at the reference coordinate **3001** and acquired in step **S507** as input, and the operating force needed to operate the hand part of the robot arm body **200** is controlled. Thus, it is possible to notify the user of a clicking sensation due to virtual attraction and/or virtual repulsion through a change in the

operating force.

[0098] Then, in step **S509**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S509** is negative, the process returns to step **S501**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S509** is affirmative, and the control flowchart ends.

[0099] According to the above-described present embodiment, the position of the robot is notified to the user by changing the operating force needed when the user directly operates the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the hand part of the robot arm body **200** while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user.

Second Embodiment

[0100] Next, the second embodiment will be described in detail. In the first embodiment, an example of the case where the direction in which the reference coordinate **3001** moves is defined as a predetermined one-axis direction (X direction) will be described. In the second embodiment, an example of the case where the direction in which the reference coordinate **3001** moves is defined as a plurality of axes, that is, X and Y-axes, X, Y, and Z-axes, or the like will be described. In teaching of a robot, a coordinate system, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, is provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate **3001** is moved in the coordinate system of the entire robot system **1000** will be described as an example. In the following description, the portions of the configuration of the hardware and control system, different from those of the first embodiment, will be described with reference to the drawings. Similar portions to those of the first embodiment have the same configuration and function as described above, and the detailed description thereof is omitted.

[0101] FIGS. **6A**, **6B**, and **6C** are diagrams for illustrating direct teaching in the present embodiment. FIG. **6A** is a diagram illustrating a virtual coordinate system, a virtual reference coordinate **3001**, and virtual force fields in a working space of the robot system **1000** in an XZ plane. FIG. **6B** is a diagram when the reference coordinate **3001** has been moved in the +X-axis direction from the state of FIG. **6A**. FIG. **6C** is a diagram when the reference coordinate **3001** has been moved in the +Z-axis direction from the state of FIG. **6B**. FIGS. **6A**, **6B**, and **6C** simply show the robot arm body **200** and the robot hand body **300** in the robot system **1000** described with reference to FIG. **1**.

[0102] From FIG. **6A**, in direct teaching, the movable range of the robot arm body **200** is limited to a predetermined direction, and furthermore, the user is notified when the hand part of the robot arm body **200** has moved a desired distance. In the present embodiment, the movable range is limited to the X-axis direction and Z-axis direction of the robot system **1000**. In other words, in the present embodiment, when the reference coordinate **3001** is moved from the state of FIG. **6A**, movement to virtual attraction coordinates **6012**, **6014**, **6017** is allowed, and movement to virtual attraction coordinates **6016**, **6018** present in the directions in the XZ plane is not allowed.

[0103] Subsequently, the reference coordinate **3001** is a reference coordinate used as a reference when virtual repulsion or attraction is acquired. When the reference coordinate **3001** is in proximity to a coordinate, at which virtual repulsion or attraction is generated, as a result of operation of the user, the operating force needed for the user to operate the hand part of the robot arm body **200** changes. The operating force changes by controlling virtual resistance at the time when the user moves the hand part of the robot arm body **200** according to the position of the reference coordinate **3001** and the virtual force field. The virtual force field is considered on the assumption

that the working space is a potential field and there are specific coordinates where repulsion or attraction is generated. When those coordinates are in close proximity to the reference coordinate **3001**, the operating force needed for the user to operate the robot arm body **200** changes under the influence of virtual repulsion or virtual attraction.

[0104] From FIG. **6A**, virtual attraction coordinates **6011**, **6012**, **6013**, **6014** are set in the X-axis direction. Furthermore, virtual attraction coordinates **6015**, **6016**, **6017**, **6018** are set on the X-axis shifted by a predetermined amount in the +Z-axis direction from the X-axis where the virtual attraction coordinates **6011**, **6012**, **6013**, **6014** are set. The state of FIG. **6A** is a state where the reference coordinate **3001** is positioned at the virtual attraction coordinate **6013**. These coordinates can be referred to as first positions or second positions.

[0105] Virtual attraction potential fields **6021**, **6022**, **6023**, **6024** are set in the X-axis direction. The virtual attraction potential field **6022** is larger than the other virtual attraction potential fields **6021**, **6023**, **6024**. Virtual repulsion potential fields **6025**, **6026** are set in the Z-axis direction. The basic matters of the virtual attraction potential field and virtual attraction range in the present embodiment are similar to those of the first embodiment, so the description is omitted.

[0106] As shown in FIG. **6B**, when the reference coordinate **3001** in the state of FIG. **6A** is moved from the virtual attraction coordinate **6013** to the virtual attraction coordinate **6014**, the reference coordinate **3001** is influenced by the virtual attraction coordinate closer to the reference coordinate **3001**. Then, in the process in which the reference coordinate **3001** approaches from the coordinate **6013** to the coordinate **6014**, the magnitude of the influence reverses, and the operating force abruptly changes. In other words, where attraction at each virtual attraction coordinate is considered as a reference, a pseudo virtual repulsion potential field is set between the virtual attraction coordinates. When the reference coordinate **3001** passes through the position where the magnitude of the influence reverses, the force pulling to the virtual attraction coordinate **6013** changes into the force pulling to the virtual attraction coordinate **6014**. In other words, the force pulling in the -X-axis direction changes into the force pushing in the +X-axis direction. Therefore, a user who is moving the hand part in the +X-axis direction can feel an abrupt change in operating force to obtain a clicking sensation.

[0107] As shown in FIG. **6C**, when the reference coordinate **3001** in the state of FIG. **6B** is moved from the virtual attraction coordinate **6014** to the virtual attraction coordinate **6018**, the reference coordinate **3001** is influenced by the virtual attraction coordinate closer to the reference coordinate **3001**. Then, in the process in which the reference coordinate **3001** approaches from the coordinate **6014** to the coordinate **6018**, the magnitude of the influence reverses, and the operating force abruptly changes. In other words, where attraction at each virtual attraction coordinate is considered as a reference, a pseudo virtual repulsion potential field is set between the virtual attraction coordinates. When the reference coordinate **3001** passes through the position where the magnitude of the influence reverses, the force pulling to the virtual attraction coordinate **6014** changes into the force pulling to the virtual attraction coordinate **6018**. In other words, the force pulling in the -Z-axis direction changes into the force pushing in the +Z-axis direction. Therefore, a user who is moving the hand part in the +Z-axis direction can feel an abrupt change in operating force to obtain a clicking sensation.

[0108] When the user feels the change in operating force, the user can grasp how many virtual attraction coordinates the reference coordinate **3001** has moved in each axis and can grasp how much the robot arm body **200** has moved. The notification of the present embodiment is to notify how much the robot arm body **200** has moved, through a change in operating force in conjunction with the movement of the position of the reference coordinate **3001**. Since a virtual attraction potential field is generated at each virtual attraction coordinate, even when the user releases his or her hand from the robot arm body **200**, the hand part of the robot arm body **200** is positioned at the virtual attraction coordinate. Therefore, it is possible to reduce the unnecessarily excessive movement of the hand part of the robot arm body **200** although the user does not intend to move

the hand, so it is preferable.

[0109] In FIG. 6A, the virtual attraction potential field **6022** is increased compared to the other virtual attraction potential fields. Thus, it is possible to make the user strongly perceive a clicking sensation for each of a predetermined number of virtual attraction coordinates, and to make the user perceive the passage of the predetermined number of virtual attraction coordinates. Therefore, it is possible to reduce the likelihood that the user erroneously counts the number of virtual attraction coordinates passed. In the present embodiment, the predetermined virtual attraction potential field in the X-axis direction is increased. Alternatively, a predetermined virtual attraction potential field in the Z-axis direction may be increased.

[0110] At the virtual attraction coordinates **6012**, **6016**, the virtual attraction potential field **6022** is set in the X-axis direction, and the virtual attraction potential fields **6025**, **6026** are set in the Z-axis direction. Although the virtual attraction potential field **6022** is larger than the virtual attraction potential fields **6025**, **6026**, all of the virtual attraction potential fields are forces that attempt to position the reference coordinate **3001** at the virtual attraction coordinate **6012** or the virtual attraction coordinate **6016**. Therefore, even when the magnitude of the potential field of a virtual force varies in the axial direction, the reference coordinate **3001** does not shift from each virtual attraction coordinate in the state where the user is not moving the hand part of the robot arm body **200**.

[0111] FIG. 7 is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. 7 is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual attraction coordinates (positions) are defined as stationary positions, and the positions where the magnitude of influence reverses are defined as notification positions. As a premise, it is assumed that virtual coordinates, virtual repulsion fields, virtual attraction fields, stationary positions, and notification positions are set in control of the robot system **1000**.

[0112] As shown in FIG. 7, initially, in step **S701**, sensor values are acquired from the encoders (not shown) provided in the motors **211** to **216**, to acquire the current position of the reference coordinate **3001**. In the present embodiment, the positions of the links **201** to **205** are acquired by the motor encoders. Alternatively, the positions of the links **201** to **205** may be directly detected by using an output shaft ENC that directly detects the positions of the links **201** to **205** or the hand part of the robot arm body **200** may be directly detected by an image capturing apparatus or the like.

[0113] Subsequently, in step **S702**, the positions where a force is applied to the reference coordinate **3001** are acquired based on the current position acquired in step **S701** and the notification positions and stationary positions set in advance. In the present embodiment, information about virtual coordinates, notification positions, stationary positions, and virtual repulsion and attraction information on virtual repulsion and attraction set at those positions is stored in a simulator storing the robot system **1000** as a model. Then, the information about the positions where a force is applied to the reference coordinate **3001** (the notification positions and the stationary positions) is acquired from the simulator. In the present embodiment, a simulator is used. Alternatively, various formats of data storing information about virtual coordinates, notification positions, stationary positions, and virtual planned repulsion and attraction set at those positions may be used.

[0114] Subsequently, in step **S703**, the force that is currently applied to the reference coordinate **3001** is acquired based on the current position of the reference coordinate **3001** and the information about the notification positions and the stationary positions. The difference from the first embodiment is that virtual force in the X-axis direction and virtual force in the Z-axis direction are taken into consideration. Subsequently, in step **S704**, the force applied by the user to the hand part of the robot arm body **200** is acquired from the values of the force sensors **251** to **256**.

[0115] Subsequently, in step **S705**, the forces in the respective axis directions (XYZ) at the reference coordinate **3001** are acquired based on the values of the force sensors **251** to **256**. In the

present embodiment, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired based on the values of the force sensors **251** to **256** and the link parameters of the robot arm body **200**. Then, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired as the forces in the respective axis directions (XYZ) at the reference coordinate **3001**. This is because the link **205** that is the distal end link as the hand part of the robot arm body **200** is easy for the user to perform direct teaching. Of course, when the robot hand body **300** is operated to perform direct teaching, the forces in the respective axis directions (XYZ) at a predetermined position in the robot hand body **300** may be acquired. In that case, the forces in the respective axis directions (XYZ) are acquired from the values of the force sensors **251** to **256**, the link parameters of the robot arm body **200**, and the shape parameters of the robot hand body **300**.

[0116] In the subsequent step **S706**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **3001** operates in the specific axis direction. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X-axis direction and the Z-axis direction at the reference coordinate **3001**. Thus, in step **S706**, the gain of the force in the Y direction of the robot arm body **200** is set to zero in the XYZ coordinate system.

[0117] In the subsequent step **S707**, the force generated at the reference coordinate **3001** is acquired based on the forces that are applied to the reference coordinate **3001** from the virtual repulsion potential fields and/or virtual attraction potential fields in each direction, set at the notification positions and the stationary positions, and the force applied from the user. Then, in step **S708**, the damping coefficient D and the spring coefficient K are changed using the force generated at the reference coordinate **3001** and acquired in step **S707** as input, and the operating force needed to operate the hand part of the robot arm body **200** is controlled. Thus, it is possible to notify the user of a clicking sensation due to virtual attraction and/or virtual repulsion through a change in the operating force.

[0118] Then, in step **S709**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S709** is negative, the process returns to step **S701**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S709** is affirmative, and the control flowchart ends.

[0119] In the present embodiment, the case where movement in the X-axis direction and Z-axis direction is allowed has been described. Alternatively, as shown in FIG. **8**, coordinates (stationary positions) where virtual attraction or repulsion is generated in the X-axis direction, Y-axis direction, and Z-axis direction may be set, and a clicking sensation may be generated in each axis direction.

[0120] The positions where the clicking sensation is generated are notification positions set for each axis as described above. It should be noted that, for the sake of simplification of illustration, FIG. **8** is not a diagram in which coordinates where attraction or repulsion is generated do not correspond in detail to the potential fields of virtual forces in each axis direction. In actual control, the coordinates (stationary positions) where virtual attraction or repulsion is generated are set so as to correspond to the potential fields of virtual forces.

[0121] According to the above-described present embodiment, the position of the robot is notified to the user by changing the operating force in a plurality of axes, needed when the user directly operates the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Third Embodiment

[0122] Next, the third embodiment will be described in detail. In the third embodiment, a mode in which the strength of the virtual force applied from a stationary position or a notification position is changed depending on the movable direction of the robot during direct teaching will be described. In teaching of a robot, coordinates, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate **3001** is moved in the coordinate system of the entire robot system **1000** will be described as an example. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted.

[0123] FIG. **9** is a diagram for illustrating direct teaching in the present embodiment. FIG. **9** is a diagram illustrating a virtual coordinate system, a virtual reference coordinate **3001**, and virtual force fields in a working space of the robot system **1000** in an XZ plane. FIG. **9** simply shows the robot arm body **200** and the robot hand body **300** in the robot system **1000** described with reference to FIG. **1**. In FIG. **9**, for the sake of ease illustration, the case of two axes, that is, the X-axis direction and the Z-axis direction, will be described; however, it is also applicable to three axes, that is, the X-axis direction, the Y-axis direction, and the Z-axis direction, as shown in FIG. **8**.

[0124] From FIG. **9**, virtual attraction coordinates **6011**, **6012**, **6013**, **6014** are set in the X-axis direction. Furthermore, virtual attraction coordinates **6015**, **6016**, **6017**, **6018** are set on the X-axis shifted by a predetermined amount in the +Z-axis direction from the X-axis where the virtual attraction coordinates **6011**, **6012**, **6013**, **6014** are set. The state of FIG. **9** is a state where the reference coordinate **3001** is positioned at the virtual attraction coordinate **6013**. These coordinates can be referred to as first positions or second positions.

[0125] Virtual attraction potential fields **7001**, **7002**, **7003**, **7004** are set in the X-axis direction. Virtual attraction potential fields **7005**, **7006** are set in the Z-axis direction. As shown in FIG. **9**, in the present embodiment, the virtual attraction potential fields set in the X-axis direction are increased compared to the virtual attraction potential fields set in the Z-axis direction. Where the virtual attraction potential field in the X-axis direction is denoted as F_x and the virtual attraction potential field in the Z-axis direction is denoted as F_z , $F_x > F_z$. In other words, when the reference coordinate **3001** is moved, the force needed to move in the X-axis direction is larger than the force needed to move in the Z-axis direction.

[0126] The virtual attraction potential fields set in the X-axis direction are different in magnitude from the virtual attraction potential fields set in the Z-axis direction. However, each of the virtual attraction potential fields is a force that attempts to position the reference coordinate **3001** at the corresponding virtual attraction coordinate. Therefore, even when the magnitude of the potential field of a virtual force varies in the axial direction, the reference coordinate **3001** does not shift from each virtual attraction coordinate in the state where the user is not moving the hand part of the robot arm body **200**. The basic matters of the virtual attraction potential field and virtual attraction range in the present embodiment are similar to those of the above-described various embodiments, so the description is omitted.

[0127] In the present embodiment described above, it is possible to move in the X-axis direction with a large force in consideration of fine adjustments between coordinates and to significantly move in the Z-axis direction with a small force. When the operating force needed for movement for each coordinate (unit) is large, a certain amount of operating force is needed even for slight movement like fine adjustments by reducing the interval between coordinates (making the interval finer). Therefore, it is possible to make fine adjustments without delicate adjustments of the applied force, and it is possible to reduce the unnecessary movement of the robot although fine adjustments

are intended, so it is suitable. On the other hand, when the operating force needed for movement for each coordinate (unit) is small, it is possible to move between coordinates with a small force, so it is possible to move with a small force also at the time of moving a long distance and it is possible to reduce burden on the user, so it is suitable. Therefore, when the operating force needed for movement is changed according to the direction to be moved, it is possible to achieve fine position adjustments and reduction of load of movement in a predetermined direction, so it is possible to improve the operability of robot operation by the user. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Fourth Embodiment

[0128] Next, the fourth embodiment will be described in detail. In the fourth embodiment, a mode in which the strength of the force applied from a stationary position or a notification position in each axis direction is switched based on the direction in which the robot arm body **200** is moved will be described. In teaching of a robot, coordinates, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate **3001** is moved in the coordinate system of the entire robot system **1000** will be described as an example. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted.

[0129] FIG. **10** is a diagram for illustrating direct teaching in the present embodiment. FIG. **10** is a diagram of the case where the reference coordinate **3001** has been moved from the state of FIG. **9** to the virtual attraction coordinate **6014**. The positions of the virtual attraction coordinates set in FIGS. **9** and **10** are the same. The present embodiment differs from the above-described various embodiments in that the virtual attraction potential fields in the X-axis direction and the virtual attraction potential fields in the Z-axis direction are switched based on the movement of the reference coordinate **3001**. FIG. **10** simply shows the robot arm body **200** and the robot hand body **300** in the robot system **1000** described with reference to FIG. **1**. In FIG. **10**, for the sake of ease illustration, the case of two axes, that is, the X-axis direction and the Z-axis direction, will be described; however, it is also applicable to three axes, that is, the X-axis direction, the Y-axis direction, and the Z-axis direction, as shown in FIG. **8**.

[0130] As shown in FIG. **10**, when the reference coordinate **3001** has been moved from the state of FIG. **9** to the virtual attraction coordinate **6014**, virtual attraction potential fields **7011**, **7012**, **7013**, **7014** are set in the X-axis direction, and virtual attraction potential fields **7015**, **7016** are set in the Z-axis direction. As shown in FIG. **10**, in the present embodiment, the virtual attraction potential fields set in the X-axis direction are reduced compared to the virtual attraction potential fields set in the Z-axis direction. Where the virtual attraction potential field in the X-axis direction is denoted as F_x and the virtual attraction potential field in the Z-axis direction is denoted as F_z , $F_z > F_x$. In the present embodiment, it is assumed that, before reaching the state of FIG. **9**, the reference coordinate **3001** is at the virtual attraction coordinate **6012** and then the reference coordinate **3001** has moved in the +X-axis direction to the virtual attraction coordinate **6013**.

[0131] In other words, in the present embodiment, when the reference coordinate **3001** moves consecutively in the axis direction in which the reference coordinate **3001** has moved immediately before (the reference coordinate **3001** moves in the same direction consecutively at least twice), the operating force needed for movement in the axis direction is switched so as to be smaller than those in the other axis directions. In the case of the present embodiment, since the reference coordinate **3001** moves at least twice in the +X-axis direction, the operating force needed for movement in the X-axis direction is reduced. This makes it easier to move in the axis direction in which the user

considers to actively move, so it is suitable. In the present embodiment, when the reference coordinate **3001** moves consecutively in the +X-axis direction (the reference coordinate **3001** moves in the same direction consecutively at least twice), the operating force is reduced. Alternatively, the operating force may be reduced when the reference coordinate **3001** moves consecutively in the -X-axis direction (the reference coordinate **3001** moves in the same direction consecutively at least twice).

[0132] The operating force may be reduced when the reference coordinate **3001** has been moved once in the +X-axis direction and then moved once in the -X-axis direction. However, when the reference coordinate **3001** is moving alternately in the + direction or in the - direction, there is a high possibility that the user is making fine adjustments, so it is preferable to reduce the operating force when the reference coordinate **3001** consecutively moves in the +X-axis direction (or -X-axis direction) (the reference coordinate **3001** moves in the same direction consecutively at least twice). The same applies in the Z-axis direction. In the case of three axes, for example, when the reference coordinate **3001** consecutively moves in the X-axis direction, where the virtual attraction potential field in the X-axis direction is denoted as F_x , the virtual attraction potential field in the Y-axis direction is defined as F_y , and the virtual attraction potential field in the Z-axis direction is denoted as F_z , it is sufficient to ensure that $F_z, F_y > F_x$.

[0133] FIG. **11** is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. **11** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual attraction coordinates (positions) are defined as stationary positions, and the positions where the magnitude of influence reverses are defined as notification positions. As a premise, it is assumed that virtual coordinates, virtual repulsion fields, virtual attraction fields, stationary positions, and notification positions are set in control of the robot system **1000**.

[0134] As shown in FIG. **11**, initially, in step **S1101**, sensor values are acquired from the encoders (not shown) provided in the motors **211** to **216**, to acquire the current position of the reference coordinate **3001**. In the present embodiment, the positions of the links **201** to **205** are acquired by the motor encoders. Alternatively, the positions of the links **201** to **205** may be directly detected by using an output shaft ENC that directly detects the positions of the links **201** to **205** or the hand part of the robot arm body **200** may be directly detected by an image capturing apparatus or the like.

[0135] Subsequently, in step **S1102**, the positions where a force is applied to the reference coordinate **3001** are acquired based on the current position acquired in step **S1101** and the notification positions and stationary positions set in advance. In the present embodiment, information about virtual coordinates, notification positions, stationary positions, and virtual repulsion and attraction information on virtual repulsion and attraction set at those positions is stored in a simulator storing the robot system **1000** as a model. Then, the information about the positions where a force is applied to the reference coordinate **3001** (the notification positions and the stationary positions) is acquired from the simulator. In the present embodiment, a simulator is used. Alternatively, various formats of data storing information about virtual coordinates, notification positions, stationary positions, and virtual planned repulsion and attraction set at those positions may be used.

[0136] Subsequently, in step **S1103**, the force that is currently applied to the reference coordinate **3001** is acquired based on the current position of the reference coordinate **3001** and the information about the notification positions and the stationary positions. In step **S1103**, virtual forces in the X-axis direction and the Z-axis direction are taken into consideration. Subsequently, in step **S1104**, the force applied by the user to the hand part of the robot arm body **200** is acquired from the values of the force sensors **251** to **256**.

[0137] Subsequently, in step **S1105**, the forces in the respective axis directions (XYZ) at the reference coordinate **3001** are acquired based on the values of the force sensors **251** to **256**. In the present embodiment, the forces in the respective axis directions (XYZ) at the predetermined

position of the link **205** are acquired based on the values of the force sensors **251** to **256** and the link parameters of the robot arm body **200**. Then, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired as the forces in the respective axis directions (XYZ) at the reference coordinate **3001**. This is because the link **205** that is the distal end link as the hand part of the robot arm body **200** is easy for the user to perform direct teaching. Of course, when the robot hand body **300** is operated to perform direct teaching, the forces in the respective axis directions (XYZ) at a predetermined position in the robot hand body **300** may be acquired. In that case, the forces in the respective axis directions (XYZ) are acquired from the values of the force sensors **251** to **256**, the link parameters of the robot arm body **200**, and the shape parameters of the robot hand body **300**.

[0138] In the subsequent step **S1106**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **3001** operates in the specific axis direction. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X-axis direction and the Z-axis direction at the reference coordinate **3001**. Thus, in step **S1106**, the gain of the force in the Y direction of the robot arm body **200** is set to zero in the XYZ coordinate system.

[0139] In the subsequent step **S1107**, the force generated at the reference coordinate **3001** is acquired based on the forces that are applied to the reference coordinate **3001** from the virtual repulsion potential fields and/or virtual attraction potential fields in each direction, set at the notification positions and the stationary positions, and the force applied from the user. Then, in step **S1108**, the damping coefficient D and the spring coefficient K are changed using the force generated at the reference coordinate **3001** and acquired in step **S1107** as input, and the operating force needed to operate the hand part of the robot arm body **200** is controlled. Thus, it is possible to notify the user of a clicking sensation due to virtual attraction and/or virtual repulsion through a change in the operating force.

[0140] Then, in step **S1109**, it is determined whether the potential distribution of the virtual attraction potential fields is changed in a specific axis direction. In the present embodiment, when the reference coordinate **3001** has moved consecutively in a specific axis direction (when the reference coordinate **3001** has moved at least twice), the magnitude of the virtual attraction potential field in the specific axis direction is made smaller than the magnitudes of the virtual attraction potential fields in the other axis directions. In the present embodiment, it is defined as at least two consecutive movements; however, any number may be set. When the determination is affirmative in **S1109**, the process proceeds to **S1110**, and the magnitude of the virtual attraction potential field in the specific axis direction is reduced compared to the virtual attraction potential fields in the other axis directions. Then, the process proceeds to step **S1111**. When the determination is negative in step **S1109**, the process proceeds to step **S1111** without changing the potential distribution of the potential fields.

[0141] Then, in step **S1111**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S1111** is negative, the process returns to step **S1101**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S1111** is affirmative, and the control flowchart ends.

[0142] According to the present embodiment described above, it is possible to easily move the robot in the axis direction in which the user considers to actively move the robot. Therefore, it is possible to reduce the burden on the user and improve the operability of robot operation by the user. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Fifth Embodiment

[0143] Next, the fifth embodiment will be described in detail. In the fifth embodiment, in the case where a predetermined joint of the robot arm body **200** is moved, the case where the amount of

movement (the amount of rotation) of the predetermined joint is notified to the user through a clicking sensation will be described. In teaching of a robot, coordinates, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, an example of the case where a rotating coordinate system at the joint J2 in movement of a reference coordinate **1201** (described later) will be described. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted.

[0144] FIG. **12** is a diagram for illustrating direct teaching in the present embodiment. FIG. **12** is a diagram illustrating a virtual coordinate system, the virtual reference coordinate **1201**, and virtual force fields in the rotation range of the joint J2 in a θ -axis direction that is the rotation direction of the joint J2. FIG. **12** simply shows the robot arm body **200** and the robot hand body **300** in the robot system **1000** described with reference to FIG. **1**. In the present embodiment, the joint J2 will be described in detail in an example; however, similar control can also be executed for the other joints. The joint J2 can be referred to as a predetermined part.

[0145] From FIG. **12**, in direct teaching, the user is notified when the joint J2 has moved a desired distance (angle). The reference coordinate **1201** is a reference coordinate used as a reference when virtual repulsion or attraction is acquired. When the reference coordinate **1201** is in proximity to a coordinate at which virtual repulsion or attraction is generated as a result of operation of the user, the operating force needed for the user to operate the link **202** connected to the joint J2 changes. The operating force changes by controlling virtual resistance at the time when the user moves the link **202** according to the position of the reference coordinate **1201** and the virtual force fields. The virtual force fields are considered on the assumption that the rotation range of the joint J2 is a potential field and there are specific coordinates where repulsion or attraction is generated. When those coordinates are in proximity to the reference coordinate **1201**, the operating force needed for the user to operate the link **202** is changed under the influence of virtual repulsion or virtual attraction.

[0146] From FIG. **12**, virtual attraction coordinates **1202** to **1213** are set in the θ -axis direction. The state of FIG. **12** is a state where the reference coordinate **1201** is positioned at the virtual attraction coordinate **1212**. From FIG. **12**, virtual attraction potential fields are set in the axis direction. For the sake of convenience of description, FIG. **12** shows virtual attraction potential fields **1221** to **1227**; however, it is assumed that corresponding virtual attraction potential fields are set at all virtual attraction coordinates **1202** to **1213**. The virtual attraction potential field **1224** and the virtual attraction potential field (not shown) set at the virtual attraction coordinate **1208** are set so as to be larger than the other virtual attraction potential fields. The basic matters of the virtual attraction potential field and virtual attraction range in the present embodiment are similar to those of the above-described various embodiments, so the description is omitted. These coordinates can be referred to as first positions or second positions.

[0147] When the reference coordinate **1201** is moved from the virtual attraction coordinate **1212** to the virtual attraction coordinate **1211**, the reference coordinate **1201** is influenced by the virtual attraction coordinate closer to the reference coordinate **1201**. Then, in the process in which the reference coordinate **1201** approaches from the virtual attraction coordinate **1212** to the virtual attraction coordinate **1211**, the magnitude of the influence reverses, and the operating force abruptly changes. In other words, where attraction at each virtual attraction coordinate is considered as a reference, a pseudo virtual repulsion potential field is set between the virtual attraction coordinates. When the reference coordinate **1201** passes through the position where the magnitude of the influence reverses, the force pulling to the virtual attraction coordinate **1212**

changes into the force pulling to the virtual attraction coordinate **1211**. In other words, the force pulling in the $+\theta$ -axis direction changes into the force pushing in the $-\theta$ -axis direction. Therefore, a user who is moving the hand part in the $-\theta$ -axis direction can feel an abrupt change in operating force to obtain a clicking sensation.

[0148] When the user feels the change in operating force, the user can grasp how many virtual attraction coordinates the reference coordinate **1201** has moved in the θ -axis direction and can grasp how much the link **202** has moved. The notification of the present embodiment is to notify how much the link **202** has moved, through a change in operating force in conjunction with the movement of the position of the reference coordinate **1201**. Since a virtual attraction potential field is generated at each virtual attraction coordinate, even when the user releases his or her hand from the robot arm body **200**, the link **202** is positioned at the virtual attraction coordinate. Therefore, it is possible to reduce the unnecessarily excessive movement of the link **202** although the user does not intend to move the hand, so it is preferable.

[0149] In FIG. **12**, the virtual attraction potential field **1224** and the virtual attraction potential field (not shown) set at the virtual attraction coordinate **1208** are set so as to be larger than the other virtual attraction potential fields. Thus, it is possible to make the user strongly perceive a clicking sensation for each of a predetermined number of virtual attraction coordinates, and to make the user perceive the passage of the predetermined number of virtual attraction coordinates. In the present embodiment, a strong clicking sensation is provided at the position that is half the rotation range of the joint J2, and it is possible for the user to intuitively grasp half rotation of the joint J2, so it is preferable. Therefore, it is possible to reduce the likelihood that the user erroneously counts the number of virtual attraction coordinates passed.

[0150] Where the joint J2 is rotatable by 360 degrees or more, when the joint J2 has been moved in the $+\theta$ direction from the state of FIG. **12** and positioned at the virtual attraction coordinate **1202**, the virtual attraction potential fields **1221** to **1124** may be used again. As shown in FIG. **13**, when the joint J2 is rotatable by 360 degrees or more, the virtual attraction potential fields after one rotation (after 360-degree rotation) are increased compared to the virtual attraction potential fields before one rotation (before 360-degree rotation). Thus, a user is able to easily grasp one rotation through the strong clicking sensation, so it is preferable. Therefore, it is possible to reduce the likelihood that the user erroneously counts the number of rotations of the joint J2.

[0151] FIG. **14** is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. **14** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual attraction coordinates (angles) are defined as stationary positions, and the positions (angles) where the magnitude of influence reverses are defined as notification positions. As a premise, it is assumed that virtual coordinates, virtual repulsion fields, virtual attraction fields, stationary positions, and notification positions are set in control of the robot system **1000**.

[0152] From FIG. **14**, initially, in step S**1401**, the sensor value is acquired from the encoder (not shown) provided in the motor **212**, to acquire the current position of the reference coordinate **1201** (rotation angle of the joint J2), thus acquiring the current position of the reference coordinate **1201**. In the present embodiment, the position of the link **202** is acquired with the motor encoder. Alternatively, an output shaft ENC that directly detects the position of the link **202** may be used, or the predetermined position of the link **202** may be directly detected by using an image capturing apparatus or the like.

[0153] Subsequently, in step S**1402**, the positions (angles) where a force is applied to the reference coordinate **1201** are acquired based on the current position acquired in step S**1401** and the notification positions and stationary positions set in advance. In the present embodiment, information about virtual coordinates, notification positions, stationary positions, and virtual repulsion and attraction information on virtual repulsion and attraction set at those positions is stored in a simulator storing the robot system **1000** as a model. Then, the information about the

positions where a force is applied to the reference coordinate **1201** (the notification positions and the stationary positions) is acquired from the simulator. In the present embodiment, a simulator is used. Alternatively, various formats of data storing information about virtual coordinates, notification positions, stationary positions, and virtual planned repulsion and attraction set at those positions may be used.

[0154] Subsequently, in step **S1403**, the force that is currently applied to the reference coordinate **1201** is acquired based on the current position of the reference coordinate **1201** and the information about the notification positions and the stationary positions. Subsequently, in step **S1404**, the force applied by the user to the link **202** is acquired from the value of the force sensor **252**.

[0155] Subsequently, in step **S1405**, the force in the θ -axis direction at the reference coordinate **1201** is acquired from the value of the force sensor **252**. In the present embodiment, the force in the θ -axis direction at the predetermined position of the link **202** is acquired based on the value of the force sensor **252** and the link parameters of the link **202**. Then, the force in the θ -axis direction at the predetermined position of the link **202** is acquired as the force in the θ -axis direction at the reference coordinate **1201**.

[0156] In the subsequent step **S1406**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **1201** operates in the specific axis direction. In the present embodiment, the link **202** is controlled to move in the θ -axis direction at the reference coordinate **1201**.

[0157] In the subsequent step **S1407**, the force generated at the reference coordinate **1201** is acquired based on the forces that are applied to the reference coordinate **1201** from the virtual repulsion potential fields and/or virtual attraction potential fields set at the notification positions and the stationary positions, and the force applied from the user.

[0158] Then, in step **S1408**, the damping coefficient D and the spring coefficient K are changed using the force generated at the reference coordinate **1201** and acquired in step **S1407** as input, and the operating force needed to operate the link **202** is controlled. Thus, it is possible to notify the user of a clicking sensation due to virtual attraction and/or virtual repulsion through a change in the operating force.

[0159] Then, in step **S1409**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S1409** is negative, the process returns to step **S1401**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S1409** is affirmative, and the control flowchart ends.

[0160] According to the above-described present embodiment, the position (angle) of the link **202** is notified to the user by changing the operating force needed when the user directly operates the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user.

[0161] In the present embodiment, the description is made by way of an example of the joint **J2**; however, the configuration is not limited thereto. For example, the operation may be performed at the joints **J1**, **J3**, **J4**, **J5**, **J6**. The control in the present embodiment may be executed for all the joints, or the control in the present embodiment may be executed at some of the joints. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Sixth Embodiment

[0162] Next, the sixth embodiment will be described in detail. In the sixth embodiment, a light **260** is illuminated as a display device (indicator) provided on the robot arm body **200** during the execution of direct teaching, to notify how much the hand part of the robot arm body **200** has

moved. In the present embodiment, a light is used; however, an apparatus that emits light as needed, such as a beacon light, may be used. In teaching of a robot, coordinates, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate **3001** is moved in the coordinate system of the entire robot system **1000** will be described as an example. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted.

[0163] FIG. **15** is a diagram in which virtual coordinates in the X-axis direction, set in the present embodiment, are set at equal intervals. In the present embodiment, for the sake of convenience of description, the movement of the hand part of the robot arm body **200** is defined in the X-axis direction; however, the movement may be defined in a plurality of axes (two axes or three axes) as in the case of the second embodiment or may be defined in the rotation axis as in the case of the fifth embodiment.

[0164] As shown in FIG. **15**, virtual coordinates **1501** to **1508** are set in the X-axis direction, and the light **260** illuminates when the reference coordinate **3001** is positioned at or passes through any one of the virtual coordinates. Then, in the state of FIG. **15**, the reference coordinate **3001** is positioned at the virtual coordinate **1505**. From this state, when the reference coordinate **3001** is moved from the virtual coordinate **1505** to the virtual coordinate **1506**, the light **260** is illuminated at the timing when the reference coordinate **3001** is positioned at the virtual coordinate **1506**. These coordinates can be referred to as first positions or second positions.

[0165] Thus, when the user feels the blinking of the light **260**, the user can grasp how many virtual coordinates the reference coordinate **3001** has moved and can grasp how much the robot arm body **200** has moved. The notification of the present embodiment is to notify how much the robot arm body **200** has moved, through blinking of light in conjunction with the movement of the position of the reference coordinate **3001**. These coordinates can be referred to as first positions or second positions.

[0166] FIG. **16** is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. **16** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual coordinates (the positions where the light **260** is blinked) are defined as notification positions. As a premise, it is assumed that virtual coordinates are set in control of the robot system **1000**.

[0167] As shown in FIG. **16**, initially, in step **S1601**, sensor values are acquired from the encoders (not shown) provided in the motors **211** to **216**, to acquire the current position of the reference coordinate **3001**. In the present embodiment, the positions of the links **201** to **205** are acquired by the motor encoders. Alternatively, the positions of the links **201** to **205** may be directly detected by using an output shaft ENC that directly detects the positions of the links **201** to **205** or the hand part of the robot arm body **200** may be directly detected by an image capturing apparatus or the like.

[0168] Subsequently in step **S1602**, the positions where light is emitted are acquired based on the current position acquired in step **S1601** and the notification positions set in advance. In the present embodiment, virtual coordinates are stored in a simulator that stores the robot system **1000** as a model. Then, the information about the positions where light is emitted (the notification positions) is acquired from the simulator. In the present embodiment, a simulator is used. Alternatively, various formats of data storing virtual coordinates may be used.

[0169] Subsequently, in step **S1603**, it is determined whether the current position of the reference coordinate **3001** is the position where light is emitted. When the determination is affirmative in step **S1603**, the process proceeds to step **S1604**, and the light **260** is blinked. When the determination is

negative in step **S1603**, the light **260** is not blinked, and the process proceeds to step **S1605**.

[0170] Subsequently, in step **S1605**, the force applied by the user to the hand part of the robot arm body **200** is acquired from the values of the force sensors **251** to **256**. Subsequently, in step **S1606**, the forces in the respective axis directions (XYZ) at the reference coordinate **3001** are acquired based on the values of the force sensors **251** to **256**. In the present embodiment, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired based on the values of the force sensors **251** to **256** and the link parameters of the robot arm body **200**. Then, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired as the forces in the respective axis directions (XYZ) at the reference coordinate **3001**. This is because the link **205** that is the distal end link as the hand part of the robot arm body **200** is easy for the user to perform direct teaching. Of course, when the robot hand body **300** is operated to perform direct teaching, the forces in the respective axis directions (XYZ) at a predetermined position in the robot hand body **300** may be acquired. In that case, the forces in the respective axis directions (XYZ) are acquired from the values of the force sensors **251** to **256**, the link parameters of the robot arm body **200**, and the shape parameters of the robot hand body **300**. [0171] In the subsequent step **S1607**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **3001** operates in the specific axis direction. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X-axis direction at the reference coordinate **3001**. Thus, in step **S506**, the gains of the forces in the Y direction and Z direction of the robot arm body **200** are set to zero in the XYZ coordinate system. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X direction. Alternatively, the hand part of the robot arm body **200** may be moved in the Y direction or may be moved in the Z direction.

[0172] In the subsequent step **S1608**, the force generated at the reference coordinate **3001** is acquired based on the force applied from the user. Then, in step **S1609**, the damping coefficient D and the spring coefficient K are changed using the force generated at the reference coordinate **3001** and acquired in step **S1608** as input, and the operating force needed to operate the hand part of the robot arm body **200** is controlled.

[0173] Then, in step **S1610**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S1610** is negative, the process returns to step **S1601**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S1610** is affirmative, and the control flowchart ends.

[0174] According to the above-described present embodiment, the position of the robot is notified to the user by emitting light when the user directly operates the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user.

[0175] When the movement is being performed consecutively in the same direction on the same axis, the blinking cycle of the light **260** may be set so as to correspond to the number of virtual coordinates passed. For example, consider the case of moving the reference coordinate **3001** from the virtual coordinate **1505** to the virtual coordinate **1508**, as shown in FIG. 17. In that case, from FIG. 17, when the reference coordinate is positioned at the virtual coordinate **1506**, the light **260** is lit once in a second (blinking at a frequency of once per second). When the reference coordinate **3001** is positioned at the virtual coordinate **1507** further from that state, the light **260** is lit twice in a second (blinking at a frequency of twice per second) because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. When the reference coordinate **3001** is positioned at the virtual coordinate **1508** further from that state, the light **260** is lit three times in a second (blinking at a frequency of three times per second) because the reference coordinate **3001**

is consecutively moving in the same direction on the same axis. In the present embodiment, the cycle is set to one second; however, a selected cycle may be set. Then, when the movement has not been detected for a predetermined time, the number of times of blinking is reset.

[0176] FIG. **18** is a control flowchart when the light **260** shown in FIG. **17** is blinked multiple times. The control flowchart described with reference to FIG. **18** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. The control flowchart described with reference to FIG. **18** includes the process of step **S1801** to step **1807** in addition to the control flowchart shown in FIG. **16**.

[0177] Initially, in step **S1801**, it is determined whether the reference coordinate **3001** is consecutively moving among the positions where light is emitted in the same direction on the same axis. When the determination is negative in step **S1801**, the process proceeds to step **S1802**, and the light **260** is blinked once as described above. Then, in step **S1803**, the number of a counter that counts the virtual coordinates that the reference coordinate **3001** has passed through is set to one. By setting the count to one, the count can always be set to one when the direction of operation (+ direction or – direction, or the X-axis direction, the Y-axis direction, or the Z-axis direction) changes.

[0178] When the determination is affirmative in step **S1801**, the process proceeds to step **S1804**, and the counter that counts the virtual coordinates that the reference coordinate **3001** has passed through is incremented by one. This is because the reference coordinate **3001** has passed through the virtual coordinates at least twice since the reference coordinate **3001** is “consecutively” moving according to the determination of step **S1801**. Then, in step **S1805**, the light **260** is blinked as described above according to the count.

[0179] Then, in step **S1806**, it is determined whether there is any operation within a predetermined time. When an operation is received from the user within the predetermined time, the process proceeds to step **S1605**. When there is no operation from the user within the predetermined time, the count is reset in step **S1807**. Thus, since the virtual coordinates are counted while the user is operating the robot, it is possible for the user to intuitively grasp how much the robot has moved since the start of the operation while the user is operating the robot.

[0180] Step **S1806** and step **S1807** may be omitted. In that case, it is possible for the user to intuitively grasp the amount of movement from when the direction of operation is changed. When the control flowchart ends, the counter is reset.

[0181] Thus, it is possible for the user to further intuitively grasp how much the hand part of the robot arm body **200** has been moved. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Seventh Embodiment

[0182] Next, the seventh embodiment will be described in detail. In the seventh embodiment, a speaker **270** provided on the robot arm body **200** produces sound during the execution of direct teaching, to notify how much the hand part of the robot arm body **200** has moved. In the present embodiment, a speaker will be described as an example; however, any apparatus that produces sound as needed may be used. In teaching of a robot, coordinates, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate **3001** is moved in the coordinate system of the entire robot system **1000** will be described as an example. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted.

[0183] FIG. **19** is a diagram in which virtual coordinates in the X-axis direction, set in the present embodiment, are set at equal intervals. In the present embodiment, for the sake of convenience of

description, the movement of the hand part of the robot arm body **200** is defined in the X-axis direction; however, the movement may be defined in a plurality of axes (two axes or three axes) as in the case of the second embodiment or may be defined in the rotation axis as in the case of the fifth embodiment.

[0184] As shown in FIG. **19**, virtual coordinates **1501** to **1508** are set in the X-axis direction, and the speaker **270** produces sound when the reference coordinate **3001** is positioned at or passes through any one of the virtual coordinates. Then, in the state of FIG. **19**, the reference coordinate **3001** is positioned at the virtual coordinate **1505**. From this state, when the reference coordinate **3001** is moved from the virtual coordinate **1505** to the virtual coordinate **1506**, the speaker **270** produces a predetermined sound at the timing when the reference coordinate **3001** is positioned at the virtual coordinate **1506**. These coordinates can be referred to as first positions or second positions.

[0185] Thus, when the user feels the sound from the speaker **270**, the user can grasp how many virtual coordinates the reference coordinate **3001** has moved and can grasp how much the robot arm body **200** has moved. The notification of the present embodiment is to notify how much the robot arm body **200** has moved, through production of sound in conjunction with the movement of the position of the reference coordinate **3001**.

[0186] FIG. **20** is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. **20** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual coordinates (the positions where sound is produced from the speaker **270**) are defined as notification positions. As a premise, it is assumed that virtual coordinates are set in control of the robot system **1000**.

[0187] As shown in FIG. **20**, initially, in step **S2001**, sensor values are acquired from the encoders (not shown) provided in the motors **211** to **216**, to acquire the current position of the reference coordinate **3001**. In the present embodiment, the positions of the links **201** to **205** are acquired by the motor encoders. Alternatively, the positions of the links **201** to **205** may be directly detected by using an output shaft ENC that directly detects the positions of the links **201** to **205** or the hand part of the robot arm body **200** may be directly detected by an image capturing apparatus or the like.

[0188] Subsequently in step **S2002**, the positions where sound is produced are acquired based on the current position acquired in step **S2001** and the notification positions set in advance. In the present embodiment, virtual coordinates are stored in a simulator that stores the robot system **1000** as a model. Then, the information about the positions where sound is produced (the notification positions) is acquired from the simulator. In the present embodiment, a simulator is used.

Alternatively, various formats of data storing virtual coordinates may be used.

[0189] Subsequently, in step **S2003**, it is determined whether the current position of the reference coordinate **3001** is the position where sound is produced. When the determination is affirmative in step **S2003**, the process proceeds to step **S2004**, and a predetermined sound is produced from the speaker **270**. The type of sound may be any one of a chime, a buzzer, and a desired sound. When the determination is negative in step **S2003**, no sound is produced from the speaker **270**, and the process proceeds to step **S2005**.

[0190] Subsequently, in step **S2005**, the force applied by the user to the hand part of the robot arm body **200** is acquired from the values of the force sensors **251** to **256**. Subsequently, in step **S2006**, the forces in the respective axis directions (XYZ) at the reference coordinate **3001** are acquired based on the values of the force sensors **251** to **256**. In the present embodiment, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired based on the values of the force sensors **251** to **256** and the link parameters of the robot arm body **200**. Then, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired as the forces in the respective axis directions (XYZ) at the reference coordinate **3001**. This is because the link **205** that is the distal end link as the hand part of the robot arm body **200** is easy for the user to perform direct teaching. Of course, when the robot hand body **300** is

operated to perform direct teaching, the forces in the respective axis directions (XYZ) at a predetermined position in the robot hand body **300** may be acquired. In that case, the forces in the respective axis directions (XYZ) are acquired from the values of the force sensors **251** to **256**, the link parameters of the robot arm body **200**, and the shape parameters of the robot hand body **300**. [0191] In the subsequent step **S2007**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **3001** operates in the specific axis direction. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X-axis direction at the reference coordinate **3001**. Thus, in step **S2006**, the gains of the forces in the Y direction and Z direction of the robot arm body **200** are set to zero in the XYZ coordinate system. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X direction. Alternatively, the hand part of the robot arm body **200** may be moved in the Y direction or may be moved in the Z direction.

[0192] In the subsequent step **S2008**, the force generated at the reference coordinate **3001** is acquired based on the force applied from the user. Then, in step **S2009**, the damping coefficient **D** and the spring coefficient **K** are changed using the force generated at the reference coordinate **3001** and acquired in step **S2008** as input, and the operating force needed to operate the hand part of the robot arm body **200** is controlled.

[0193] Then, in step **S2010**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S2010** is negative, the process returns to step **S2001**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S2010** is affirmative, and the control flowchart ends.

[0194] According to the above-described present embodiment, the position of the robot is notified to the user by producing sound when the user directly operates the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user.

[0195] When the movement is being performed consecutively in the same direction on the same axis, sound produced from the speaker **270** may be set so as to correspond to the number of virtual coordinates passed. For example, consider the case of moving the reference coordinate **3001** from the virtual coordinate **1505** to the virtual coordinate **1508**, as shown in FIG. **21**. In that case, from FIG. **21**, when the reference coordinate is positioned at the virtual coordinate **1506**, the speaker **270** is caused to pronounce “one”. When the reference coordinate **3001** is positioned at the virtual coordinate **1507** further from that state, the speaker **270** is caused to pronounce “two” because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. When the reference coordinate **3001** is positioned at the virtual coordinate **1508** further from that state, the speaker **270** is caused to pronounce “three” because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. Then, when the movement has not been detected for a predetermined time, the number of times of count is reset.

[0196] FIG. **22** is a control flowchart when the corresponding sound is produced by the speaker **270** shown in FIG. **21**. The control flowchart described with reference to FIG. **22** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. The control flowchart described with reference to FIG. **22** includes the process of step **S2201** to step **2207** in addition to the control flowchart shown in FIG. **20**.

[0197] Initially, in step **S2201**, it is determined whether the reference coordinate **3001** is consecutively moving among the positions where sound is produced in the same direction on the same axis. When the determination is negative in step **S2201**, the process proceeds to step **S2202**, and the speaker **270** pronounces “one” as described above. Then, in step **S2203**, the number of a counter that counts the virtual coordinates that the reference coordinate **3001** has passed through is

set to one. By setting the count to one, the count can always be set to one when the direction of operation (+ direction or – direction, or the X-axis direction, the Y-axis direction, or the Z-axis side) changes.

[0198] When the determination is affirmative in step **S2201**, the process proceeds to step **S2204**, and the counter that counts the virtual coordinates that the reference coordinate **3001** has passed through is incremented by one. This is because the reference coordinate **3001** has passed through the virtual coordinates at least twice since the reference coordinate **3001** is “consecutively” moving according to the determination of step **S2201**. Then, in step **S2205**, the speaker **270** is caused to pronounce a sound corresponding to the count as described above.

[0199] Then, in step **S2206**, it is determined whether there is any operation within a predetermined time. When an operation is received from the user within the predetermined time, the process proceeds to step **S2005**. When there is no operation from the user within the predetermined time, the process proceeds to step **S2207**, and the count is reset. Thus, since the virtual coordinates are counted while the user is operating the robot, it is possible for the user to intuitively grasp how much the robot has moved since the start of the operation while the user is operating the robot. Step **S2206** and step **S2207** may be omitted. In that case, it is possible for the user to intuitively grasp the amount of movement from when the direction of operation is changed. When the control flowchart ends, the counter is reset.

[0200] When the movement is being performed consecutively in the same direction on the same axis, sound from the speaker **270** may be consecutively produced so as to correspond to the number of virtual coordinates passed. For example, consider the case of moving the reference coordinate **3001** from the virtual coordinate **1505** to the virtual coordinate **1508**, as shown in FIG. 23. In that case, from FIG. 23, when the reference coordinate is positioned at the virtual coordinate **1506**, a buzzer sound is produced once in a second from the speaker **270** (producing a buzzer sound at a frequency of once per second). When the reference coordinate **3001** is positioned at the virtual coordinate **1507** further from that state, the speaker **270** is caused to produce a buzzer sound twice in a second (produce a buzzer sound at a frequency of twice per second) because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. When the reference coordinate **3001** is positioned at the virtual coordinate **1508** further from that state, the speaker **270** is caused to produce a buzzer sound three times in a second (produce a buzzer sound at a frequency of three times per second) because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. In the present embodiment, the cycle is set to one second; however, a selected cycle may be set. Then, when the movement has not been detected for a predetermined time, the number of times of blinking is reset.

[0201] Thus, it is possible for the user to further intuitively grasp how much the hand part of the robot arm body **200** has been moved. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification. For example, by combining the present embodiment with the first to fifth embodiments in which notification is provided through a change in operating force as described above, a clicking sensation is generated at the notification position, and a clicking sound is also produced. Thus, it is possible to further clearly notify the user of the clicking sensation and the notification position, so it is also possible to improve an operating feel on the robot.

Eighth Embodiment

[0202] Next, the eighth embodiment will be described in detail. In the eighth embodiment, numeric values are displayed on the display portion **280** serving as a display on the link **205** of the hand part of the robot arm body **200** during the execution of direct teaching, to notify how much the hand part of the robot arm body **200** has moved. In teaching of a robot, coordinates, such as a robot coordinate system, a tool coordinate system, and a user coordinate system, are provided according to the application; however, any coordinate system is applicable, and another coordinate for this teaching may be provided. In the present embodiment, the case where the reference coordinate

3001 is moved in the coordinate system of the entire robot system **1000** will be described as an example. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted.

[0203] FIG. **24** is a diagram in which virtual coordinates in the X-axis direction, set in the present embodiment, are set at equal intervals. In the present embodiment, for the sake of convenience of description, the movement of the hand part of the robot arm body **200** is defined in the X-axis direction; however, the movement may be defined in a plurality of axes (two axes or three axes) as in the case of the second embodiment or may be defined in the rotation axis as in the case of the fifth embodiment.

[0204] As shown in FIG. **24**, virtual coordinates **1501** to **1508** are set in the X-axis direction, and a corresponding numeric value is displayed on the display portion **280** when the reference coordinate **3001** is positioned at or passes through any one of the virtual coordinates. Then, in the state of FIG. **24**, the reference coordinate **3001** is positioned at the virtual coordinate **1505**. From this state, when the reference coordinate **3001** is moved from the virtual coordinate **1505** to the virtual coordinate **1506**, the corresponding numeric value is displayed on the display portion **280** at the timing when the reference coordinate **3001** is positioned at the virtual coordinate **1506**. These coordinates can be referred to as first positions or second positions.

[0205] In the present embodiment, when the movement is being performed consecutively in the same direction on the same axis, the numeric value to be displayed on the display portion **280** may be set so as to correspond to the number of virtual coordinates passed. Consider the case of moving the reference coordinate **3001** from the virtual coordinate **1505** to the virtual coordinate **1508**, as shown in FIG. **24**. In that case, from FIG. **24**, when the reference coordinate is positioned at the virtual coordinate **1506**, the display portion **280** is caused to display “one”. When the reference coordinate **3001** is positioned at the virtual coordinate **1507** further from that state, the display portion **280** is caused to display “two” because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. When the reference coordinate **3001** is positioned at the virtual coordinate **1508** further from that state, the display portion **280** is caused to display “three” because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. Then, when the movement has not been detected for a predetermined time, the number of times of count is reset.

[0206] Thus, when the user visually recognizes the display from the display portion **280**, the user can grasp how many virtual coordinates the reference coordinate **3001** has moved and can grasp how much the robot arm body **200** has moved. The display portion **280** is provided at the hand part of the robot arm body **200**, so visibility for the user improves. The notification of the present embodiment is to notify how much the robot arm body **200** has moved, through display of the numeric value in conjunction with the movement of the position of the reference coordinate **3001**.

[0207] FIG. **25** is a control flowchart in the present embodiment. The control flowchart described with reference to FIG. **25** is executed through the coordination between the CPU **401** of the controller **400** and the CPU mounted at each joint. Here, the virtual coordinates (the positions where the numeric value is displayed on the display portion **280**) are defined as notification positions. As a premise, it is assumed that virtual coordinates are set in control of the robot system **1000**. As a premise, the initial display of the display portion **280** is set such that the display portion **280** displays a numeric value of “zero” or does not display a numeric value.

[0208] As shown in FIG. **25**, initially, in step S2501, sensor values are acquired from the encoders (not shown) provided in the motors **211** to **216**, to acquire the current position of the reference coordinate **3001**. In the present embodiment, the positions of the links **201** to **205** are acquired by the motor encoders. Alternatively, the positions of the links **201** to **205** may be directly detected by

using an output shaft ENC that directly detects the positions of the links **201** to **205** or the hand part of the robot arm body **200** may be directly detected by an image capturing apparatus or the like. [0209] Subsequently, in step **S2502**, the positions where the numeric value is updated are acquired based on the current position acquired in step **S2501** and the notification positions set in advance. In the present embodiment, virtual coordinates are stored in a simulator that stores the robot system **1000** as a model. Then, the information about the positions where the numeric value is updated (the notification positions) is acquired from the simulator. In the present embodiment, a simulator is used. Alternatively, various formats of data storing virtual coordinates may be used.

[0210] Subsequently, in step **S2503**, it is determined whether the current position of the reference coordinate **3001** is the position where the numeric value is updated. When the determination is affirmative in step **S2503**, the process proceeds to step **S2504**. When the determination is negative in step **S2503**, the display portion **280** does not update the display of the numeric value, and the process proceeds to step **S2511**.

[0211] Subsequently, in step **S2504**, it is determined whether the reference coordinate **3001** is moving among the positions where the display of the numeric value is updated in the same direction on the same axis. When the determination is negative in step **S2504**, the process proceeds to step **S2505**, and the display portion **280** shows “one” as described above. Then, in step **S2506**, the number of a counter that counts the virtual coordinates that the reference coordinate **3001** has passed through is set to one. By setting the count to one, the count can always be set to one when the direction of operation (+ direction or – direction, or the X-axis direction, the Y-axis direction, or the Z-axis side) changes.

[0212] When the determination is affirmative in step **S2504**, the process proceeds to step **S2507**, and the counter that counts the virtual coordinates that the reference coordinate **3001** has passed through is incremented by one. This is because the reference coordinate **3001** has passed through the virtual coordinates at least twice since the reference coordinate **3001** is “consecutively” moving according to the determination of step **S2504**. Then, in step **S2508**, the display portion **280** displays the numeric value corresponding to the count as described above.

[0213] Then, in step **S2509**, it is determined whether there is any operation within a predetermined time. When an operation is received from the user within the predetermined time, the process proceeds to step **S2512**. When there is no operation from the user within the predetermined time, the process proceeds to step **S2510**, the count is reset, and, in step **S2511**, the display portion **280** displays a numeric value of “zero” or does not display the numeric value. Thus, since the virtual coordinates are counted while the user is operating the robot, it is possible for the user to intuitively grasp how much the robot has moved since the start of the operation while the user is operating the robot. Step **S2509** and step **S2510** may be omitted. In that case, it is possible for the user to intuitively grasp the amount of movement from when the direction of operation is changed. When the control flowchart ends, the counter is reset.

[0214] Subsequently, in step **S2512**, the force applied by the user to the hand part of the robot arm body **200** is acquired from the values of the force sensors **251** to **256**. Subsequently, in step **S2513**, the forces in the respective axis directions (XYZ) at the reference coordinate **3001** are acquired based on the values of the force sensors **251** to **256**. In the present embodiment, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired based on the values of the force sensors **251** to **256** and the link parameters of the robot arm body **200**. Then, the forces in the respective axis directions (XYZ) at the predetermined position of the link **205** are acquired as the forces in the respective axis directions (XYZ) at the reference coordinate **3001**. This is because the link **205** that is the distal end link as the hand part of the robot arm body **200** is easy for the user to perform direct teaching. Of course, when the robot hand body **300** is operated to perform direct teaching, the forces in the respective axis directions (XYZ) at a predetermined position in the robot hand body **300** may be acquired. In that case, the forces in the respective axis directions (XYZ) are acquired from the values of the force sensors **251** to **256**, the

link parameters of the robot arm body **200**, and the shape parameters of the robot hand body **300**.
[0215] In the subsequent step **S2514**, gains of the forces in directions other than a specific axis direction are set to zero such that the reference coordinate **3001** operates in the specific axis direction. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X-axis direction at the reference coordinate **3001**. Thus, in step **S2514**, the gains of the forces in the Y direction and Z direction of the robot arm body **200** are set to zero in the XYZ coordinate system. In the present embodiment, the hand part of the robot arm body **200** is controlled to move in the X direction. Alternatively, the hand part of the robot arm body **200** may be moved in the Y direction or may be moved in the Z direction.

[0216] In the subsequent step **S2515**, the force generated at the reference coordinate **3001** is acquired based on the force applied from the user. Then, in step **S2516**, the damping coefficient **D** and the spring coefficient **K** are changed using the force generated at the reference coordinate **3001** and acquired in step **S2515** as input, and the operating force needed to operate the hand part of the robot arm body **200** is controlled.

[0217] Then, in step **S2517**, it is determined whether the user has provided instructions to end direct teaching. When the user has not provided instructions to end direct teaching, the determination in step **S2517** is negative, the process returns to step **S2501**, and the above-described control flowchart is repeated. When the user has provided instructions to end direct teaching, the determination in step **S2517** is affirmative, and the control flowchart ends.

[0218] According to the above-described present embodiment, the position of the robot is notified to the user by displaying a numeric value when the user directly operates the robot. The numeric value is displayed at the hand part of the robot arm body **200**. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Ninth Embodiment

[0219] Next, the ninth embodiment will be described in detail. In the ninth embodiment, the user interface for setting the conditions of the case where the above-described various embodiments are implemented will be described. In the following description, the portions of the configuration of the hardware and control system, different from those of the above-described various embodiments, will be described with reference to the drawings. Similar portions to those of the above-described various embodiments have the same configuration and function as described above, and the detailed description thereof is omitted. The robot system **1000** of the present embodiment is configured such that the above-described various embodiments can be implemented.

[0220] FIG. **26** shows a direct teaching settings screen **800** according to the present embodiment. When the direct teaching settings screen **800** is displayed in a touch panel format on the display portion **280** provided at the hand part of the robot arm body **200**, the direct teaching settings screen **800** allows the user to easily change settings during direct teaching, so it is suitable. The direct teaching settings screen **800** may be displayed on the external input device **500**, or may be displayed on a desktop PC, a laptop PC, a tablet PC, a smartphone, and the like, that are general-purpose computers capable of performing information processing.

[0221] From FIG. **26**, initially, a hand part Cartesian coordinate system button **801** (radio button) and a joint coordinate system button **802** (radio button), with which it is possible to set or select virtual coordinates in which coordinate system, are shown on the direct teaching settings screen **800**. In the example of FIG. **26**, the hand part Cartesian coordinate system button **801** is selected. When the hand part Cartesian coordinate system button **801** (radio button) is selected, input of position interval input boxes **803**, **804**, **805** (described later) becomes active, and the radio buttons

of the corresponding axes are in a selected state. When the joint coordinate system button **802** is selected, input of position interval input boxes **806, 807, 808, 809, 810, 811** (described later) becomes active, and the radio buttons of the corresponding axes are in a selected state. When both the hand part Cartesian coordinate system button **801** and the joint coordinate system button **802** are selected, input of all the position interval input boxes becomes active, and the radio buttons of the corresponding axes are in a selected state.

[0222] The position interval input box **803** allows to input the interval at which virtual coordinates are set in the X-axis direction in which the reference coordinate set at the hand part of the robot arm body **200** moves. By selecting the radio button on the sheet left side of the position interval input box **803**, movement in the X-axis direction is allowed. When the radio button is deselected, movement in the X-axis direction is disabled. The position interval input box **804** allows to input the interval at which virtual coordinates are set in the Y-axis direction in which the reference coordinate set at the hand part of the robot arm body **200** moves. By selecting the radio button on the sheet left side of the position interval input box **804**, movement in the Y-axis direction is allowed. When the radio button is deselected, movement in the Y-axis direction is disabled. The position interval input box **805** allows to input the interval at which virtual coordinates are set in the Z-axis direction in which the reference coordinate set at the hand part of the robot arm body **200** moves. By selecting the radio button on the sheet left side of the position interval input box **805**, movement in the Z-axis direction is allowed. When the radio button is deselected, movement in the Z-axis direction is disabled.

[0223] The position interval input box **806** allows to input the interval at which the virtual coordinates are set in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J1 moves. By selecting the radio button on the sheet left side of the position interval input box **806**, movement in the θ -axis direction (rotation direction) of the joint J1 is allowed. The position interval input box **807** allows to input the interval at which the virtual coordinates are set in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J2 moves. By selecting the radio button on the sheet left side of the position interval input box **807**, movement in the θ -axis direction (rotation direction) of the joint J2 is allowed. The position interval input box **808** allows to input the interval at which the virtual coordinates are set in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J3 moves. By selecting the radio button on the sheet left side of the position interval input box **808**, movement in the θ -axis direction (rotation direction) of the joint J3 is allowed.

[0224] The position interval input box **809** allows to input the interval at which the virtual coordinates are set in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J4 moves. By selecting the radio button on the sheet left side of the position interval input box **809**, movement in the θ -axis direction (rotation direction) of the joint J4 is allowed. The position interval input box **810** allows to input the interval at which the virtual coordinates are set in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J5 moves. By selecting the radio button on the sheet left side of the position interval input box **810**, movement in the θ -axis direction (rotation direction) of the joint J5 is allowed. The position interval input box **811** allows to input the interval at which the virtual coordinates are set in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J6 moves. By selecting the radio button on the sheet left side of the position interval input box **811**, movement in the θ -axis direction (rotation direction) of the joint J6 is allowed.

[0225] By inputting numeric values to the position interval input boxes **803** to **805**, virtual coordinates can be set at intervals in meters. In the present embodiment, the setting is in meters. Alternatively, the setting may be in millimeters or centimeters. By inputting numeric values to the position interval input boxes **806** to **811**, virtual coordinates can be set at intervals in degrees ($^{\circ}$). In the present embodiment, the setting is in degrees ($^{\circ}$). Alternatively, the setting may be in radians. A numeric value may be input to the position interval input boxes **803** to **811** by directly inputting a

displayed value with a keyboard and a mouse or setting a value with the up-down arrow buttons of each of the position interval boxes **803** to **811** using the mouse or touch.

[0226] From FIG. **26**, the direct teaching settings screen **800** shows buttons for setting a mode in which the amount of movement of the robot arm body **200** is notified to the user. Control to provide notification through a change in operating force as described in the first to fifth embodiments is executed with an operating force notification mode button **812** (radio button). Control to provide notification through emitting light as described in the sixth embodiment is executed with a light notification mode button **813** (radio button). Control to provide notification through producing sound as described in the seventh embodiment is executed with a sound notification mode button **814** (radio button). Control to provide notification through updating the display of a numeric value as described in the eighth embodiment is executed with a numerical notification mode button **815** (radio button). All selected modes may be configured to be executed, or only one mode may be configured to be selected, or only two or three modes may be configured to be selected.

[0227] Subsequently, the direct teaching settings screen **800** shows the notification mode advanced settings screen **850** for advanced settings in the selected notification mode. The virtual repulsion setting input box **816** allows to set the maximum value in newtons of the virtual repulsion potential field uniformly set at each of the set virtual coordinates. A virtual repulsion range and a virtual repulsion potential distribution are set according to the input. The virtual attraction setting input box **817** allows to set the maximum value in newtons of the virtual attraction potential field uniformly set at each of the set virtual coordinates. A virtual attraction range and a virtual attraction potential distribution are set according to the input. It is assumed that one of the values is controlled to be zero. For example, when a value is input to the virtual repulsion setting input box **816**, the virtual attraction setting input box **817** automatically becomes zero; whereas, when a value is input to the virtual attraction setting input box **817**, the virtual repulsion setting input box **816** automatically becomes zero. A numeric value may be input to the virtual repulsion setting input box **816** and the virtual attraction setting input box **817** by directly inputting a displayed value with a keyboard and a mouse or setting a value with the up-down arrow buttons of each input box using the mouse or touch. In the present embodiment, the setting is in newtons. Alternatively, the setting in kgf may be implemented.

[0228] A one rotation notification mode button **818** (radio button) is a button for notifying the user of one rotation when multiple rotations (360° or more) are possible at the predetermined joint of the robot arm body **200**, as described with reference to the fifth embodiment. When the one rotation notification mode button **818** is selected, the virtual repulsion potential field or attraction potential field after one rotation is increased compared to that before the one rotation. In the present embodiment, when the one rotation notification mode button **818** is selected, the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** is automatically set to 200% of the input value. However, an input box may be prepared as needed to allow input of a value to be increased or a multiplication factor from the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** may be configured to be set.

[0229] A half rotation notification mode button **819** (radio button) is a button for notifying the user of a rotation up to half of the range (half rotation) in the rotation range of the predetermined joint of the robot arm body **200** as described in the fifth embodiment. When the half rotation notification mode button **819** is selected, the repulsion potential field or attraction potential field of a virtual coordinate positioned halfway in the rotation range of the predetermined joint of the robot arm body **200** or near the position is increased. In the present embodiment, when the half rotation notification mode button **819** is selected, the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** is automatically set to 200% of the input value. However, an input box may be prepared as needed to allow input of a value to be increased or a

multiplication factor from the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** may be configured to be set.

[0230] A consecutive-operation operating force change mode button **820** (radio button) is a button of a mode for changing the potential distribution of the force fields of the axis when movement is consecutive in the same axis direction, as described in the fourth embodiment. In the present embodiment, when the consecutive-operation operating force change mode button **820** is selected, the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** is automatically set to 50% of the input value. However, an input box may be prepared as needed to allow input of a value to be set after change or a reduction factor from the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** may be configured to be set.

[0231] An off-site avoidance and interference avoidance repulsion setting mode button **821** is a button of a mode in which repulsion potential fields are automatically set at predetermined virtual coordinate positions such that the hand part of the robot arm body **200** is not placed off the site or does not interfere with surrounding objects as described in the first embodiment. Whether the hand part of the robot arm body **200** is off the site and avoidance of interference with surrounding objects are determined by using a robot model set in advance by a robot simulator, and repulsion potential fields are automatically set at predetermined virtual coordinate positions.

[0232] A predetermined number enhanced notification mode button **822** is a button of a mode in which the virtual attraction potential field or virtual repulsion potential field is enhanced every predetermined number of virtual coordinates as described in the above embodiments. A force field setting input box **823** allows to set how many times to multiply the force field, at which enhanced notification is performed, by the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817**. A numeric value may be input to the force field setting input box **823** by directly inputting a displayed value with a keyboard and a mouse or setting a value with the up-down arrow buttons of the input box using the mouse or touch. In the present embodiment, the multiplication factor is set. Alternatively, a percentage (%) may be set.

[0233] A number input box **824** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the X-axis direction in which the reference coordinate set at the hand part of the robot arm body **200** moves. When the radio button on the sheet left side of the number input box **824** is selected, the enhanced notification mode is set in the X-axis direction. When the radio button is deselected, the enhanced notification mode is not set in the X-axis direction. A number input box **825** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the Y-axis direction in which the reference coordinate set at the hand part of the robot arm body **200** moves. When the radio button on the sheet left side of the number input box **825** is selected, the enhanced notification mode is set in the Y-axis direction. When the radio button is deselected, the enhanced notification mode is not set in the Y-axis direction. A number input box **826** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the Z-axis direction in which the reference coordinate set at the hand part of the robot arm body **200** moves. When the radio button on the sheet left side of the number input box **826** is selected, the enhanced notification mode is set in the Z-axis direction. When the radio button is deselected, the enhanced notification mode is not set in the Z-axis direction.

[0234] A number input box **827** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint **J1** moves. When the radio button on the sheet left side of the number input box **827** is selected, the enhanced notification mode is set in the θ -axis direction (rotation direction) of the joint **J1**. When the radio button is deselected, the enhanced notification mode is not set in the θ -axis direction (rotation direction) of the joint **J1**. A number input box **828** allows to input the interval at which the repulsion potential field or attraction

potential field is enhanced at virtual coordinates in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J2 moves. When the radio button on the sheet left side of the number input box **828** is selected, the enhanced notification mode is set in the θ -axis direction (rotation direction) of the joint J2. When the radio button is deselected, the enhanced notification mode is not set in the θ -axis direction (rotation direction) of the joint J2. A number input box **829** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J3 moves. When the radio button on the sheet left side of the number input box **829** is selected, the enhanced notification mode is set in the θ -axis direction (rotation direction) of the joint J3. When the radio button is deselected, the enhanced notification mode is not set in the θ -axis direction (rotation direction) of the joint J3.

[0235] A number input box **830** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J4 moves. When the radio button on the sheet left side of the number input box **830** is selected, the enhanced notification mode is set in the θ -axis direction (rotation direction) of the joint J4. When the radio button is deselected, the enhanced notification mode is not set in the θ -axis direction (rotation direction) of the joint J4. A number input box **831** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J5 moves. When the radio button on the sheet left side of the number input box **831** is selected, the enhanced notification mode is set in the θ -axis direction (rotation direction) of the joint J5. When the radio button is deselected, the enhanced notification mode is not set in the θ -axis direction (rotation direction) of the joint J5. A number input box **832** allows to input the interval at which the repulsion potential field or attraction potential field is enhanced at virtual coordinates in the θ -axis direction (rotation direction) in which the reference coordinate set at the joint J6 moves. When the radio button on the sheet left side of the number input box **832** is selected, the enhanced notification mode is set in the θ -axis direction (rotation direction) of the joint J6. When the radio button is deselected, the enhanced notification mode is not set in the θ -axis direction (rotation direction) of the joint J6.

[0236] A numeric value may be input to the number input boxes **824** to **832** by directly inputting a displayed value with a keyboard and a mouse or setting a value with the up-down arrow buttons of each of the number input boxes **824** to **832** using the mouse or touch.

[0237] An X-direction enhancement button **837** (radio button) is a button of a mode in which the virtual repulsion potential fields or virtual attraction potential fields in the X-axis direction in which the reference coordinate of the hand part of the robot arm body **200** moves are increased as described in the third embodiment. A Y-direction enhancement button **838** (radio button) is a mode in which the virtual repulsion potential fields or virtual attraction potential fields in the Y-axis direction in which the reference coordinate of the hand part of the robot arm body **200** moves are increased as described in the third embodiment. A Z-direction enhancement button **839** (radio button) is a mode in which the virtual repulsion potential fields or virtual attraction potential fields in the Z-axis direction in which the reference coordinate of the hand part of the robot arm body **200** moves are increased as described in the third embodiment. In the present embodiment, when the X-direction enhancement button **837**, the Y-direction enhancement button **838**, or the Z-direction enhancement button **839** is selected, the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** is automatically set to 200% of the input value.

However, an input box may be prepared one by one as needed to allow input of a value to be increased or a multiplication factor from the value input to the virtual repulsion setting input box **816** or the virtual attraction setting input box **817** may be configured to be set.

[0238] A light cycle setting input box **833** allows to set the cycle for blinking the light **260** in correspondence with the virtual coordinates passed, as described in the sixth embodiment. A sound

cycle setting input box **834** allows to set the cycle for producing sound from the speaker **270** in correspondence with the virtual coordinates passed, as described in the seventh embodiment. A numeric value may be input to the light cycle setting input box **833** and the sound cycle setting input box **834** by directly inputting a value displayed with a keyboard and a mouse. A numeric value may be input by setting a value with the up-down arrow buttons of each of the number input boxes **824** to **832** using the mouse or touch. In the present embodiment, a numeric value is input to the light cycle setting input box **833** and the sound cycle setting input box **834** in seconds(s). Alternatively, a numeric value may be input in milliseconds (ms) or minutes.

[0239] A sound setting box **835** allows to select the type of sound, as described in the seventh embodiment. By using the up and down arrow buttons of the sound setting box **835**, various sounds set in advance, such as buzzer, chime, and alarm, can be selected. A reset time setting input box **836** allows to set the duration of a non-operated state of the user when the count of virtual coordinates passed is reset, as described in the sixth to eighth embodiments. A numeric value may be input to the reset time setting input box **836** by directly inputting a value displayed with a keyboard and a mouse. A numeric value may be input by setting a value with the up-down arrow button of the reset time setting input box **836** using the mouse or touch. In the present embodiment, a numeric value may be input to the reset time setting input box **836** in seconds(s). Alternatively, a numeric value may be input in milliseconds (ms) or minutes.

[0240] According to the present embodiment, it is possible to easily perform settings for notifying the user of the amount of movement of the robot. Thus, for example, it is possible to make the user intuitively grasp the position of the robot according to user's feeling while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification.

Other Embodiments

[0241] In the above-described sixth to eighth embodiments, notifications corresponding to virtual coordinates are made using light, sound, numerical display, and the like. However, it is also applicable to notify the user through vibration, other than light, sound, or numerical display. For example, a vibrator **290** is provided on the link **205** of the robot arm body **200** as shown in FIG. 27. Then, notification may be made by vibrating the vibrator **290** when the reference coordinate **3001** is positioned at or passes through a virtual coordinate.

[0242] When the movement is being performed consecutively in the same direction on the same axis, the vibrator **290** may be consecutively vibrated in correspondence with the number of virtual coordinates passed. For example, consider the case of moving the reference coordinate **3001** from the virtual coordinate **1505** to the virtual coordinate **1508**, as shown in FIG. 27. These coordinates can be referred to as first positions or second positions. In that case, from FIG. 27, when the reference coordinate is positioned at the virtual coordinate **1506**, the vibrator **290** is vibrated once in a second (vibrated at a frequency of once per second). When the reference coordinate **3001** is positioned at the virtual coordinate **1507** further from that state, the vibrator **290** is vibrated twice in a second (vibrated at a frequency of twice per second) because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. When the reference coordinate **3001** is positioned at the virtual coordinate **1508** further from that state, the vibrator **290** is vibrated three times in a second (vibrated at a frequency of three times per second) because the reference coordinate **3001** is consecutively moving in the same direction on the same axis. In the present embodiment, the cycle is set to one second; however, a selected cycle may be set. Then, when the movement has not been detected for a predetermined time, the number of times of blinking is reset. The control flowchart is similar to those of the sixth to eighth embodiments.

[0243] The above-described various embodiments and modifications may be implemented in combination with the present embodiment (notification by vibration) and/or the present modification (notification by vibration). In the direct teaching settings screen **800** of the ninth

embodiment, a vibration notification mode button and a vibration cycle setting input box may be displayed.

[0244] According to the above-described present embodiment (notification by vibration), the position of the robot is notified to the user by generating vibration when the user directly operates the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot while the user is performing direct teaching, with the result that it is possible to improve the operability of robot operation by the user. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification. For example, by combining the present embodiment with the first to fifth embodiments in which notification is provided through a change in operating force as described above, a clicking sensation is generated at the notification position, and vibration is also generated. Thus, it is possible to further clearly notify the user of the clicking sensation and the notification position, so it is also possible to improve an operating feel on the robot.

[0245] In the above-described various embodiments, an example of the case where the robot is operated through direct teaching has been described. Alternatively, for example, it is also applicable to the case where the robot is operated with an external input device (operating device), such as a teaching pendant. FIG. **28** is a diagram illustrating the case where the above-described various embodiments are implemented by using the tablet-form external input device **500**. In FIG. **28**, the description will be made by using a tablet form; however, a teaching pendant for operating the robot with a physical user interface, such as a smartphone, a general button, and a joystick, may be used.

[0246] As shown in FIG. **28**, the tablet-form external input device **500** includes a touch panel display portion **500a**. The display portion **500a** displays a virtual robot system **1000V** that represents the robot system **1000** in a virtual model, and also displays a virtual robot arm body **200V** corresponding to the robot arm body **200**, and a virtual robot hand body **300V**. By operating the virtual robot arm body **200V** and/or the virtual robot hand body **300V** while touching them with the user's finger, it is possible to operate the robot arm body **200** and/or the robot hand body **300** of the real machine. For the sake of illustrative purposes, FIG. **28** shows an enlarged view of the external input device **500** and the user's finger.

[0247] The external input device **500** includes a light **500b** capable of emitting light, a speaker **500c** capable of producing sound, and a vibrator **500d** capable of vibrating the external input device **500**. The display portion **500a** shows the virtual coordinates **1501** to **1508** and the virtual reference coordinate **3001**, and also shows a numerical display section **500e** displaying the moving distance of the reference coordinate **3001** in numeric value. Virtual force fields as described in the first to fourth embodiments are set at the virtual coordinates **1501** to **1508**. Of course, as in the case of the fifth embodiment, the display portion **500a** may show a virtual joint model and show force fields set within the rotation range at a virtual joint. In FIG. **28**, the external input device **500** shows the X-axis and allows operation on the X-axis. Alternatively, the external input device **500** may show a plurality of axes (Y-axis and Z-axis) and may allow operation on the plurality of axes (Y-axis and Z-axis).

[0248] Then, from the state of FIG. **28**, when the virtual robot hand body **300V** is moved to the virtual coordinate **1506** by the user's finger, the operating force at the time of moving the virtual robot hand body **300V** in a virtual space is controlled as in the case of the first to fourth embodiments. Thus, the user is able to perceive a virtual clicking sensation in the robot hand body **300**.

[0249] When the virtual robot hand body **300V** is moved, light from the light **500b** may be blinked as in the case of the sixth embodiment, and light may be blinked multiple times according to the amount of movement of the reference coordinate **3001**. When the virtual robot hand body **300V** is

moved, sound may be produced from the speaker **500c** as in the case of the seventh embodiment. It is possible to produce sound multiple times and produce a corresponding sound (produce the sound “one” and produce the sound “two”) according to the amount of movement of the reference coordinate **3001**. By performing a virtual clicking sensation through a virtual force field and a clicking sound from the speaker **500c** in combination, it is possible for the user to further perceive a clicking sensation, so it is possible to improve an operating feel.

[0250] When the virtual robot hand body **300V** is moved, display of the numeric value at the numerical display section **500e** may be updated as in the case of the eighth embodiment, and numerical display may be performed according to the amount of movement of the reference coordinate **3001**. When the virtual robot hand body **300V** is moved, vibration may be generated from the vibrator **500d**, and vibration may be generated multiple times according to the amount of movement of the reference coordinate **3001**. By performing a virtual clicking sensation through a virtual force field and vibration of the vibrator **500d** in combination, it is possible for the user to further perceive a clicking sensation, so it is possible to improve an operating feel. Furthermore, a virtual clicking sensation due to a virtual force field, vibration of the vibrator **500d**, and sound produced from the speaker **500c** all may be performed in combination. A virtual clicking sensation due to a virtual force field, blinking of light from the light **500b**, sound produced from the speaker **500c**, vibration of the vibrator **500d**, numerical display by the numerical display section **500e** all may be performed in combination.

[0251] According to the present embodiment (notification from the external input device), when a user operates the robot with the external input device **500**, various notifications are provided by the external input device **500** to notify the user of the position of the robot. Thus, the user is able to operate the robot while grasping the coordinate during operation with the external input device **500** (while grasping how much the robot arm body **200** or the robot hand body **300** has moved). Thus, for example, it is possible to make the user intuitively grasp the position of the robot when the user is operating the robot with the external input device **500**, with the result that it is possible to improve the operability of robot operation by the user.

[0252] By displaying virtual coordinates on the display portion **500a**, it is possible to make the user intuitively grasp how much the robot can move. Of course, in cases where notification, such as operating force, light, and sound, is provided, the virtual coordinates may be hidden, and a user interface that allows the user to select whether to display or hide the virtual coordinates may be provided on the display portion **500a** or the external input device **500**. The above-described various embodiments and modifications may be implemented in combination with the present embodiment and/or the present modification. For example, a user interface for setting the details of notification as described in the ninth embodiment may be shown on the display portion **500a**.

[0253] Procedures of the above-described embodiments are specifically executed by a CPU. Therefore, the CPU may be configured to read a recording medium, on which a program of software capable of executing the above-described functions is recorded, and run the program. In this case, the program itself read from the recording medium implements the functions of the above-described embodiments, and the program itself and the recording medium on which the program is recorded are components of the present disclosure.

[0254] In each of the embodiments, a case where a computer-readable recording medium is an ROM or an RAM or a flash ROM and a program is stored in the ROM or the RAM or the flash ROM has been described. However, the present disclosure is not limited to such modes. The program for carrying out the present disclosure may be recorded on any recording medium as long as the recording medium is a computer-readable recording medium.

[0255] In the above-described various embodiments, a case where an articulated robot arm in which the robot arm body **200** includes a plurality of joints is used has been described; however, the number of joints is not limited thereto. A vertical multi-axis configuration is described as a form of the robot arm. A configuration equivalent to the above can be implemented even with joints in a

different form, such as a horizontal articulated type, a parallel link type, and an orthogonal robot.

[0256] The above-described various embodiments are applicable to machines capable of automatically performing extension and contraction, bending and stretching, up and down movements, right and left movements, or swing operation, or a combined operation of them in accordance with information of a storage device provided in a controller.

[0257] The present disclosure is not limited to the above-described embodiments and many modifications are applicable within the technical concept of the present disclosure. Advantageous effects described in the embodiments of the present disclosure are only the most favorable advantageous effects obtained from the present disclosure, and advantageous effects of the present disclosure are not limited to those described in the embodiments of the present disclosure. The above-described various embodiments and modifications may be implemented in combination.

[0258] According to the present disclosure, it is possible to improve the operability of robot operation by a user.

Other Embodiments

[0259] Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0260] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the present disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A robot system comprising: a robot; and a controller that controls the robot, wherein the controller when a user is moving a predetermined part of the robot or moving the predetermined part with a terminal device, notifies the user that the predetermined part is moving from a first position to a second position.
2. The robot system according to claim 1, wherein virtual repulsion or virtual attraction is set at the first position and the second position, and the controller notifies the user by, based on the repulsion or the attraction, changing operating force used when the user moves the predetermined part.
3. The robot system according to claim 2, wherein the attraction is set at the first position and the second position, and the controller when the predetermined part is moving from the first position to the second position, changes the operating force based on a fact that a force set for the predetermined part changes from a force pulling to the first position to a force pulling to the second

position.

4. The robot system according to claim 2, wherein the controller generates a clicking sensation due to a change in the operating force when the user is moving the predetermined part.
5. The robot system according to claim 2, wherein the controller increases the repulsion or the attraction at every predetermined number.
6. The robot system according to claim 2, wherein the controller when the predetermined part has moved at least twice consecutively in a predetermined direction, reduces the repulsion or the attraction, set in the predetermined direction.
7. The robot system according to claim 2, wherein the controller displays, on a display portion, a screen for setting the repulsion or the attraction.
8. The robot system according to claim 5, wherein the controller displays a screen for setting the predetermined number.
9. The robot system according to claim 5, wherein the controller displays, on a display portion, a screen for setting how to increase the repulsion or the attraction at every predetermined number.
10. The robot system according to claim 6, wherein the controller when the predetermined part has moved at least twice consecutively in a predetermined direction, displays, on a display portion, a screen for setting execution of control to reduce the repulsion or the attraction, set in the predetermined direction.
11. The robot system according to claim 1, wherein the controller notifies the user by light that the predetermined part is moving from the first position to the second position.
12. The robot system according to claim 11, wherein the controller when the predetermined part has moved at least twice consecutively in a predetermined direction, blinks the light according to the number of positions of the consecutive movements.
13. The robot system according to claim 11, wherein the controller displays, on a display portion, a screen for setting information related to the notification of light.
14. The robot system according to claim 1, wherein the controller notifies the user by sound that the predetermined part is moving from the first position to the second position.
15. The robot system according to claim 14, wherein the controller when the predetermined part has moved at least twice consecutively in a predetermined direction, generates the sound according to the number of positions of the consecutive movements or repeatedly generates the sound according to the number of the consecutive movements.
16. The robot system according to claim 14, wherein the controller displays, on a display portion, a screen for setting information related to the notification of sound.
17. The robot system according to claim 1, wherein the controller notifies the user by numerical display that the predetermined part is moving from the first position to the second position.
18. The robot system according to claim 17, wherein the controller when the predetermined part has moved at least twice consecutively in a predetermined direction, displays a numeric value according to the number of positions of the consecutive movements.
19. The robot system according to claim 1, wherein the controller notifies the user by vibration that the predetermined part is moving from the first position to the second position.
20. The robot system according to claim 19, wherein the controller when the predetermined part has moved at least twice consecutively in a predetermined direction, changes the number of the vibrations generated according to the number of positions of the consecutive movements.
21. The robot system according to claim 19, wherein the controller displays, on a display portion, a screen for setting information related to the notification of vibration.
22. The robot system according to claim 1, wherein the controller displays, on a display portion, a screen for setting an interval between the first position and the second position.
23. The robot system according to claim 1, wherein displays, on a display portion, a screen for setting a mode to notify the user.
24. The robot system according to claim 1, wherein the predetermined part is a part that moves in a

three-dimensional space where the robot is placed, and the first position and the second position are coordinates virtually set on at least one axis of the space.

25. The robot system according to claim 24, wherein the controller notifies the user when the predetermined part has moved between the coordinates.

26. The robot system according to claim 1, wherein the predetermined part is a part that moves by a joint of the robot, and the first position and the second position are coordinates virtually set in a moving range of the joint.

27. The robot system according to claim 26, wherein the controller notifies the user that the joint has rotated by a half rotation.

28. The robot system according to claim 26, wherein the joint is capable of rotating by 360 degrees or more, and the controller notifies the user that the joint has rotated by 360 degrees or more.

29. The robot system according to claim 1, wherein the controller limits directions in which the user moves the predetermined part.

30. The robot system according to claim 1, further comprising an operating device enabling the robot to be operated by input from the user, wherein the controller notifies the user by the operating device that the predetermined part is moving from the first position to the second position.

31. The robot system according to claim 30, wherein a virtual robot that allows the user to be operated is displayed on a display portion of the operating device.

32. A manufacturing method for an article, comprising manufacturing an article by using the robot system according to claim 1.

33. A control method for a robot system including a robot and a controller that controls the robot, the control method comprising by the controller, when a user is moving a predetermined part of the robot or moving the predetermined part with a terminal device, notifying the user that the predetermined part is moving from a first position to a second position.

34. An information processing apparatus that sets information related to a robot system including a robot and a controller that controls the robot, wherein the information processing apparatus displays, on a display portion, a screen for setting a condition that the controller, when a user is moving a predetermined part of the robot or moving the predetermined part with a terminal device, notifies the user that the predetermined part is moving from a first position to a second position.

35. An information processing method of setting information related to a robot system including a robot and a controller that controls the robot, the information processing method comprising displaying, on a display portion, a screen for setting a condition that the controller, when a user is moving a predetermined part of the robot or moving the predetermined part with a terminal device, notifies the user that the predetermined part is moving from a first position to a second position.

36. (canceled)

37. A non-transitory computer-readable recording medium storing instructions that, when executed, configure an information processing apparatus to perform a method according to claim 35.
